Who Gets to Live in Productive Cities? Family Transfers, Housing Costs, and Spatial Sorting

October 26, 2026

Abstract

High house prices in productive cities often exceed what young adults can afford on their own. The "bank of mom and dad" can therefore influence who becomes a homeowner in these cities. This paper studies how intergenerational transfers shape young adults' location and housing-tenure choices and examines their macroeconomic implications. I develop a spatial equilibrium model with endogenous transfers, calibrated to urban China. Relative to an equilibrium without inter vivos transfers, parental support raises the share of households aged 20–29 living in the productive region by 3.5 percent and the share owning a home there by 46.0 percent. The residence response is relatively small because renting provides access to productive cities without purchasing, while modest transfers make ownership in cheaper cities more attractive. Transfers expand the productive region's workforce, but lower average worker productivity partly offsets the output contribution of this expansion. Together with the modest residence response, this limits the aggregate-output gain to 0.23 percent. Transfers also raise house prices in the productive region by 4.55 percent, increasing purchase costs for households without parental support. Transfers narrow ownership gaps across children's earning capacities but widen them by parental wealth. Experiments with public housing-affordability policies show that credit expansion partly substitutes for family support in home purchase, and rental assistance in productive-city access.

Keywords: spatial sorting; housing tenure; intergenerational transfers; family finance; housing affordability; China.

JEL Classification: D64; E21; J61; R21; R23.

1 Introduction

High-productivity cities offer higher wages, stronger career prospects, and access to denser labor markets, while their high housing costs can restrict workers' access to these opportunities and generate aggregate productivity losses (Hsieh and Moretti, 2019). Young workers are particularly constrained because they typically have limited liquid wealth to meet large down-payment requirements. When their labor income and accumulated savings are insufficient to cover these upfront costs, parental transfers can bridge the financing gap. Such support may shape where young adults live and work, whether they become homeowners, and how much wealth they accumulate over the life cycle.

Family support for housing is widespread across countries. In the United States, parental financial assistance helps young adults meet down-payment requirements and enter homeownership (Lee et al., 2020). Intergenerational support is also an important source of housing finance in China, where housing prices in major cities are high relative to household earnings (Chen and Wen, 2017, Fang et al., 2016, Meng et al., 2023b). Across these settings, parental resources supplement young households' own income and savings at a stage when housing decisions require substantial financial commitments.

Evidence from urban China highlights a distinction between residence in productive cities and homeownership within them. Using the 2018 China Household Income Project (CHIP) survey, I find that among young households living in major cities, recipients of parental housing assistance are more likely to own their homes, yet recipients are not more concentrated in these cities overall. This raises a question about family finance: how much does it expand access to productive locations, relative to its effects on homeownership and housing wealth?

In this paper, I explore how parental transfers shape young households' location choices, homeownership, and subsequent wealth accumulation. I trace the implications for equilibrium housing prices and rents, aggregate productivity, and the distribution of wealth and welfare. The aggregate consequences of parental support are ambiguous. Transfers can enable financially constrained households to enter productive locations or purchase housing after entering. Transfer-financed demand can also raise purchase prices and disadvantage prospective buyers who receive little or no family support. Moreover, when rental housing is available, an inability to buy does not necessarily exclude a household from the productive city: a young adult who cannot finance ownership may continue to live and work there as a renter. Modest transfers also make ownership in a lower-cost city more attractive, drawing some households toward buying there rather

than renting in a productive city. The effects of family finance therefore depend on households' joint location and tenure choices and the resulting adjustment of housing markets.

To study these mechanisms, I develop a six-stage overlapping-generations model with two regions that differ in productivity and housing costs. Region A represents large, high-productivity cities, while Region B offers lower earnings and more affordable housing. Households differ in talent, geographic origins, family resources, and their financial and housing positions. They choose their region of residence, housing tenure and size, liquid saving, and mortgage borrowing. Altruistic parents aged 50–59 choose a one-time transfer to their young-adult children aged 20–29 in response to the children's earning capacity and housing opportunities, subject to their own resource constraints. The model additionally incorporates mortgage amortization, home-equity accumulation, age-dependent migration costs, terminal estates, and stochastic inheritance. Purchase prices and rents are determined in equilibrium, with separate owner and rental markets whose relative supplies respond to market conditions.

The quantitative analysis uses urban Chinese data from the China Household Finance Survey (CHFS), CHIP, and related housing-market sources to discipline earnings heterogeneity, regional housing costs, age-specific location–tenure profiles, and parental-transfer receipt among young homeowners. I quantify the role of parental support by comparing the benchmark with a counterfactual stationary equilibrium without inter vivos transfers. I evaluate mortgage-credit expansion and rental assistance with and without family transfers to assess whether public support substitutes for family finance in home purchase and access to productive locations.

The main finding is that family transfers expand young households' ownership in the productive region substantially more than their residence there. Among households aged 20–29, transfers increase the share living in Region A by 3.5 percent (1.18 percentage points) and the share owning a home there by 46.0 percent (4.30 percentage points), relative to the no-transfer equilibrium. An aggregate decomposition attributes approximately 91 percent of the ownership response to the change in ownership conditional on Region A residence. Rental access allows young households to live and work in Region A without owning there, so transfers can change tenure without changing location. Meanwhile, modest transfers induce some households who would otherwise rent in A to buy in lower-cost Region B, offsetting part of the increase in A residence.

The increase in Region A residence is concentrated among financially constrained households with lower talent. These households expand the region's workforce but also shift its composition toward workers with lower effective productivity. Average effec-

tive productivity therefore falls, partly offsetting the gain in effective labor from a larger workforce. Together with the modest net increase in Region A residence, this composition effect limits the aggregate-output gain to 0.23 percent relative to the no-transfer equilibrium.

Earlier ownership strengthens household balance sheets and improves welfare. Transfers raise mean home equity among households aged 20–29 by RMB 83,400 and reduce their share with nonpositive net worth by 55.9 percent, or 28.99 percentage points, relative to the no-transfer equilibrium. Young households also hold more liquid assets and less mortgage debt. These initial differences carry into subsequent stages through saving and mortgage repayment. The no-transfer economy requires a uniform 11.60-percent increase in consumption and housing services to attain benchmark welfare. The largest welfare gains from family transfers accrue to young and lower-talent households whose housing choices are most sensitive to liquid resources.

The distributional effects of family transfers differ across children's talent and parental wealth: transfers narrow homeownership gaps across talent levels while widening them across parental-wealth groups. By easing financing constraints, they help low- and middle-talent young adults buy homes and narrow their ownership gap with high-talent households. Transfers also reduce home-equity inequality among young households. Wealthier parents, however, are more likely to provide support and give larger amounts, giving their children an advantage in buying homes and building housing equity. In Region A, the ownership gap between children in the highest and lowest parental-wealth quartiles widens from 0.52 percentage points without transfers to 10.88 percentage points with transfers.

Nonrecipients are also affected by family support through equilibrium purchase prices and rents. Transfer-supported households shift toward ownership, putting upward pressure on purchase prices and reducing pressure on the rental market. In Region A, the purchase price is 4.55 percent higher in the benchmark than in the no-transfer equilibrium, while rent is slightly lower. Prospective buyers without parental assistance consequently face higher purchase costs and larger required down payments. In the benchmark, nonrecipients who rent in Region A face slightly lower rents per square meter, while existing owners hold more valuable housing assets.

The policy experiments show that relaxing the loan-to-value (LTV) ceiling partly substitutes for parental support at purchase entry, while family resources continue to shape mortgage debt and home equity. Rental assistance reduces dependence on parental support for residence in Region A while leaving its role in ownership largely intact.

The paper contributes to three strands of research. The first examines housing costs,

worker mobility, and the spatial allocation of labor (Behrens et al., 2014, Hsieh and Moretti, 2019, Bryan and Morten, 2019). Closely related quantitative studies evaluate housing-affordability policies and the regional effects of mortgage-credit conditions (Favilukis et al., 2023, Mabille, 2023). Research on China also links housing costs and collateral constraints to household migration (Garriga et al., 2023, Meng et al., 2023a, He et al., 2026). I introduce endogenous parental transfers as an additional source of finance for young households' migration and housing purchases. Parents choose support in response to their adult children's earning capacity and location–tenure opportunities, connecting family resources to the allocation of workers across regions and to the equilibrium incidence of housing and place-based policies.

Second, the paper extends research on parental assistance and homeownership among young households. Empirical studies examine how transfers affect purchase timing, down payments, and the size and financing of the dwelling acquired (Engelhardt and Mayer, 1998, Guiso and Jappelli, 2002, Lee et al., 2020, Meng et al., 2023b). Brandsaas (2025) studies endogenous parental transfers and housing-policy effects in a life-cycle framework. I introduce adult location choice across regions with different productivity and housing costs, making the spatial sorting of labor an outcome of intergenerational finance. The spatial setting allows me to assess how family finance affects aggregate productivity and the distribution of housing wealth and welfare across regions and households.

Third, the paper relates to the broader literature on intergenerational resources, economic opportunity, and wealth inequality. This literature examines how family background shapes education, occupational choice, and adult economic outcomes, and how transfers and bequests transmit wealth across generations (De Nardi, 2004, Nishiyama, 2002, Benhabib et al., 2011, Adermon et al., 2018, Abbott et al., 2019, Boar and Lashkari, 2021, Biasi, 2023). Related housing research studies the transmission of parental housing wealth to children's homeownership and wealth, as well as the effects of local housing markets on children's education and future income (Daysal et al., 2023, Gilraine et al., 2023). Fogli et al. (2026) show how parents' neighborhood and educational choices generate a feedback between residential segregation and income inequality. Chyn and Daruich (2025) evaluate housing vouchers and place-based subsidies in a general-equilibrium model with neighborhood effects, endogenous location choice, and child development. I examine direct monetary support when adult children enter labor and housing markets and trace its consequences for their subsequent balance sheets. The framework provides a housing-finance channel through which parental resources shape adult economic status and permits a joint assessment of disparities associated with children's earning capacity

and those associated with family background.

Section 2 presents descriptive evidence on parental housing assistance, homeownership, and location. Section 3 presents the model, and Section 4 describes the calibration. Section 5 characterizes the benchmark equilibrium. Section 6 quantifies the general-equilibrium effects of inter vivos family transfers. Section 7 evaluates LTV relaxation and rental assistance. Section 8 concludes.

2 Parental Housing Assistance, Homeownership, and Location

I use the 2018 China Household Income Project (CHIP) survey to document the relationship between parental housing assistance and young households' location and tenure choices. The analysis combines the urban household and person files and focuses on 2,954 households whose heads are aged 20–39. The housing module asks whether the household received parental assistance when buying or building a dwelling. I compare households reporting any assistance with those reporting none. To distinguish location from tenure, I group the sampled urban cities into two regions. Region A groups major cities characterized by higher productivity and higher housing costs. In CHIP, this group consists of 13 cities: Beijing, Guangzhou, Shenzhen, Chengdu, Wuhan, Chongqing, Nanjing, Suzhou, Changsha, Shenyang, Qingdao, Zhengzhou, and Dalian. Region B comprises the remaining sampled urban cities, which generally have lower productivity and more affordable housing.

Restricting attention to ages 20–39 reduces the influence of China's earlier housing-allocation system. These household heads were no older than 19 when the 1998 reform began ending in-kind urban housing allocation,[1] so their adult purchase and financing decisions are more closely associated with the development of the commercial housing market. Older cohorts' current tenure may reflect work-unit allocations or collectively provided housing.

Parental housing assistance is common among these young households. Table 1 shows that 49.4 percent report receiving assistance. Among current homeowners, this share reaches 54.7 percent. Family support therefore features prominently in young households' housing finance. Its association with their housing choices, however, differs sharply between ownership and residence.

[1] See the State Council's 1998 housing-reform notice, Guo Fa No. 23, Section II.

Table 1: Prevalence of Parental Housing Assistance among Young Households

Sample	Households	Reporting assistance	Share (%)
All sampled young households	2,954	1,458	49.4
Current homeowners	2,351	1,285	54.7

Notes: CHIP 2018 urban households with heads aged 20–39. Current homeowners are a subset of the first row. Assistance is retrospective parental support for buying or building housing; ownership refers to the current dwelling. Neither sample identifies recent or first-time buyers. Source: CHIP 2018 urban household and person files; author's calculations.

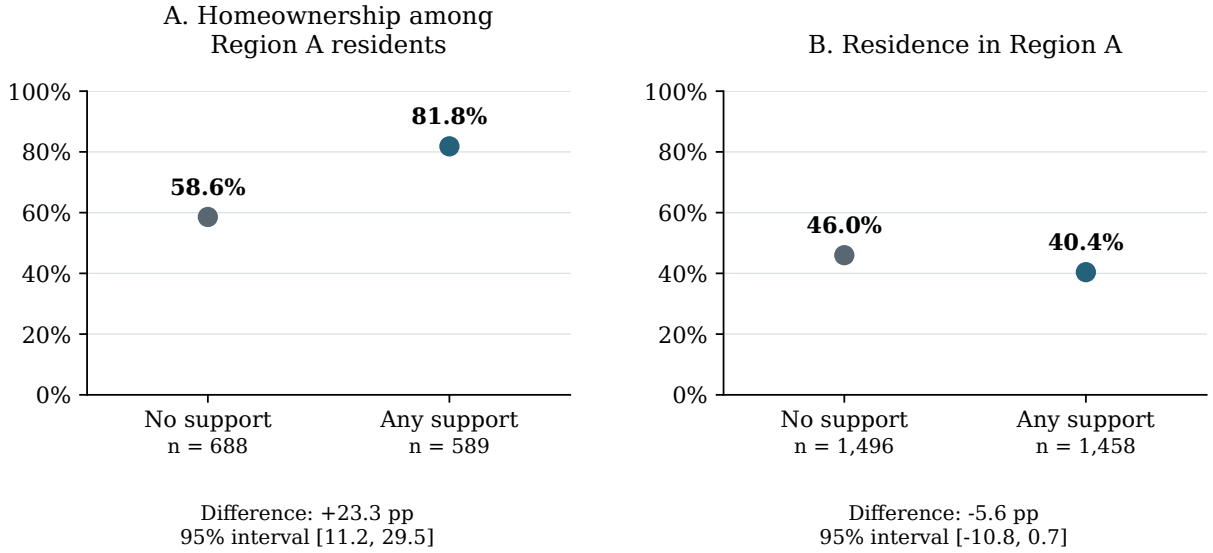
Fact 1: Recipients have higher homeownership within Region A. Panel A of Figure 1 reports homeownership conditional on residing in Region A. The ownership rate is 81.8 percent among households reporting parental housing assistance, compared with 58.6 percent among those reporting none—a gap of 23.3 percentage points. Rental tenure displays the opposite pattern: among Region A residents, 14.6 percent of recipients rent their dwelling, compared with 36.0 percent of nonrecipients. Parental housing assistance is thus associated with a pronounced difference in tenure within the productive region.

Fact 2: Recipients have a lower observed Region A residence share. Panel B reports the share residing in Region A among all households in each assistance group. This share is 40.4 percent for recipients and 46.0 percent for nonrecipients, a difference of -5.6 percentage points. The point estimates therefore show a positive association with homeownership within A and a negative association with residence there. Nevertheless, the joint share owning and residing in A is higher among recipients: 33.1 percent versus 26.9 percent. Their substantially higher ownership rate within A more than offsets their lower A residence share in this joint outcome.

Discussion. The contrast between the two panels suggests that the role of family finance depends on the alternatives available to young households. Purchasing in a productive city is one option; renting there and purchasing in a less expensive city are two others. Three mechanisms can generate different ownership and residence patterns.

First, parents may endogenously choose whether to provide a transfer based on their children's earning capacity and financing needs. Children with weaker earning capacity have fewer resources to finance a home, which may induce their parents to provide assistance. These children may also be less inclined to live in A: when earnings depend on individual productivity, weaker earning capacity reduces the additional income available from working in A, making its higher housing costs harder to justify. Recipients can

Figure 1: Parental Housing Assistance, Homeownership, and Region A Residence



Notes: CHIP 2018 urban households with heads aged 20–39 and valid parental-housing-assistance and tenure responses. Panel A reports ownership of the current dwelling conditional on Region A residence; Panel B reports Region A residence among all households in each assistance group. Assistance refers to reported historical parental support for buying or building housing. Group sample sizes are displayed. Intervals refer to recipient-minus-nonrecipient differences and use an approximate city bootstrap with 1,999 replications across 154 cities; they are not group-mean confidence intervals or full survey-design inference. Free and other accommodation remain in the residence and conditional-ownership denominators. Source: CHIP 2018 urban household and person files; author's calculations.

therefore be disproportionately drawn from households that would prefer B even without assistance. Their lower observed A residence share need not mean that assistance itself discourages residence in A; it may partly reflect whom parents choose to support.

Second, renting separates access to a productive labor market from the upfront financing required for homeownership. A young household can have sufficient earnings to pay rent in A while lacking the resources needed for a purchase. Parental assistance can then make ownership feasible without changing the household's preferred location. The large rental share among nonrecipients residing in A makes this margin particularly relevant. Through this mechanism, family finance can have a substantial effect on tenure even when its effect on residence is modest.

Third, a transfer expands the option of buying in B as well as buying in A. For a household choosing between renting in A and owning in B, assistance may make the less expensive purchase especially attractive. It can be sufficient to finance ownership in B while remaining insufficient to support the preferred purchase in A. Other households may use assistance to purchase in A instead. These competing adjustments imply that greater purchasing capacity need not translate into a greater propensity to live in the

productive region. The location response depends on which location–tenure alternative becomes more attractive, not simply on whether the household has more resources.

These mechanisms have different implications for the effect of parental assistance, yet the descriptive comparisons cannot establish their relative importance. A model is therefore needed to evaluate these mechanisms jointly and assess how much parental assistance changes where households live, relative to its effect on homeownership.

3 A Spatial Equilibrium Model with Family Transfers

This section develops a six-stage overlapping-generations model with two regions that incorporates household location and tenure choice, mortgage borrowing, inter vivos family transfers, and stochastic inheritance, with house prices and rents determined in equilibrium. Region A represents large cities with higher total factor productivity (TFP) but higher housing costs. Region B represents smaller, less productive cities where lower housing costs make homeownership more affordable. This two-region structure abstracts from the full geography of many cities, but it captures the central trade-off between productive labor-market opportunities and housing affordability.

The economy is populated by overlapping generations of heterogeneous households that differ in talent and family resources. A model period spans ten years, and each household lives for six periods, corresponding to ages 20–29, 30–39, 40–49, 50–59, 60–69, and 70–79. Households supply labor inelastically from ages 20–69 and retire at ages 70–79. The family link becomes active when a household enters the 50–59 age group and is paired with a child aged 20–29. After receiving its inheritance and observing the child's talent, the altruistic parent chooses a monetary transfer. The child inherits its origin region from the parent's beginning-of-period location, begins adult life without housing or mortgage debt, and makes an independent location and tenure choice. Households exit at the end of the 70–79 stage and leave terminal net estates. These estates determine the common inheritance distribution described below. There is no mortality before the terminal age. Population is stationary, with equal mass in each age group and total population normalized to one.

At each age, a household selects a location–tenure alternative: renting or owning in Region A or Region B. An incumbent homeowner may instead keep the existing house. Conditional on this discrete choice, the household chooses consumption, housing size, next-period liquid assets, and, when purchasing a house, mortgage borrowing. These decisions depend on talent, the household's current balance sheet, housing prices and rents, and any family transfer or inheritance received. Home purchases must satisfy the

household budget and loan-to-value and payment-to-income limits. Family resources can therefore affect both tenure and location by helping finance a down payment or entry into the high-cost Region A.

3.1 Preferences

Let $g \in \mathcal{G} \equiv \{1, \ldots, 6\}$ index ten-year age groups, ordered from ages 20–29 ($g = 1$) to ages 70–79 ($g = 6$). Let $j \in \mathcal{J} \equiv \{rB, oB, rA, oA\}$ index location–tenure choices, where r and o denote renting and owning and A and B denote the two regions. Incumbent homeowners may also keep their current house.

Conditional on consumption c, housing services h, and alternative j, the deterministic flow utility of an age-group-g household is

$$u^g(c, h, j) = \frac{\mathcal{C}(c, h)^{1-\gamma}}{1-\gamma} + \xi_j^g, \tag{1}$$

where the consumption–housing composite is

$$\mathcal{C}(c, h) = \left((1-\alpha_h)c^\epsilon + \alpha_h h^\epsilon\right)^{\frac{1}{\epsilon}}. \tag{2}$$

The parameter γ controls curvature, α_h is the weight on housing, and the elasticity of substitution between consumption and housing is $1/(1-\epsilon)$. For each age group g, the alternative-specific constants are collected in

$$\boldsymbol{\xi}^g \equiv (\xi_{rB}^g, \xi_{oB}^g, \xi_{rA}^g, \xi_{oA}^g) = (0, \xi_{oB}^g, \xi_{rA}^g, \xi_{oA}^g). \tag{3}$$

Only relative utility levels are identified, so renting in Region B is normalized to zero. The remaining terms are deterministic, age-specific residual utility shifters. After accounting for consumption and housing services, prices, household states, Region A entry costs, recurring rental frictions, and continuation values, they absorb the average nonpecuniary attractiveness of each location–tenure combination, including unmodeled amenities, tenure preferences, and location-specific ownership considerations. They are common within each age group, vary freely across age groups, and are disciplined by the corresponding age-specific location–tenure shares. In particular, ξ_{oA}^g directly captures the residual attractiveness of owning in Region A compared with renting in Region B. A homeowner who keeps the current house receives ξ_{oB}^g in Region B or ξ_{oA}^g in Region A.

In each period, a household draws an idiosyncratic taste shock ν_j for each feasible location–tenure alternative. These shocks are independently drawn from a Type-I extreme-value distribution with scale σ_ν.

Households in two age groups have additional intergenerational utility terms. In age group $g = 4$ (50–59), the parent places weight α_p on the value of its linked child, evaluated at the family transfer chosen by the parent; hence, α_p governs the strength of the altruistic transfer motive. In age group $g = 6$ (70–79), the household has a warm-glow bequest motive over its terminal net estate e:

$$\omega_b \frac{(e + \underline{e})^{1-\gamma}}{1-\gamma}, \tag{4}$$

where ω_b governs the strength of the warm-glow motive and $\underline{e} > 0$ is a small shift. Terminal estates also determine the stochastic inheritance distribution described below.

3.2 Endowments, Income and Intergenerational Links

A household enters the labor market with idiosyncratic productivity z, which remains fixed over its life cycle. As a benchmark, child productivity is drawn independently of parental productivity (Galor and Tsiddon, 1997, Benhabib et al., 2011, Colas et al., 2021). Appendix D.3 relaxes this assumption by allowing positive intergenerational persistence in talent.

During working ages $g = 1, \ldots, 5$, gross labor income in region ℓ is $y^g_\ell(z) = w\,\mathrm{TFP}_\ell G_g z$, where w is the monetary wage per productivity-adjusted efficiency unit over a ten-year model period. The term G_g captures the life-cycle productivity profile, and TFP_ℓ captures regional productivity; I normalize $\mathrm{TFP}_B = 1$. Retirement income is $y^{\mathrm{pen}}(z) = \rho_{\mathrm{pen}} w G_4 z$. The pension base $wG_4 z$ excludes regional productivity differences, so payments do not depend on the household's past or current location. Pension payments enter government expenditure and are financed from labor-tax revenue.

Within each linked parent–child pair, the altruistic parent chooses a one-time monetary transfer $f \geq 0$ in response to the child's earning capacity and housing opportunities, subject to its own resource constraints. The transfer increases the entrant's resources one-for-one and reduces the parent's resources by the same amount.

Inheritance is drawn once from the common stationary distribution of terminal estates, independently of the recipient's productivity and earlier transfer receipt. Appendix A.3 describes its construction. Figure 2 summarizes the life-cycle timing.

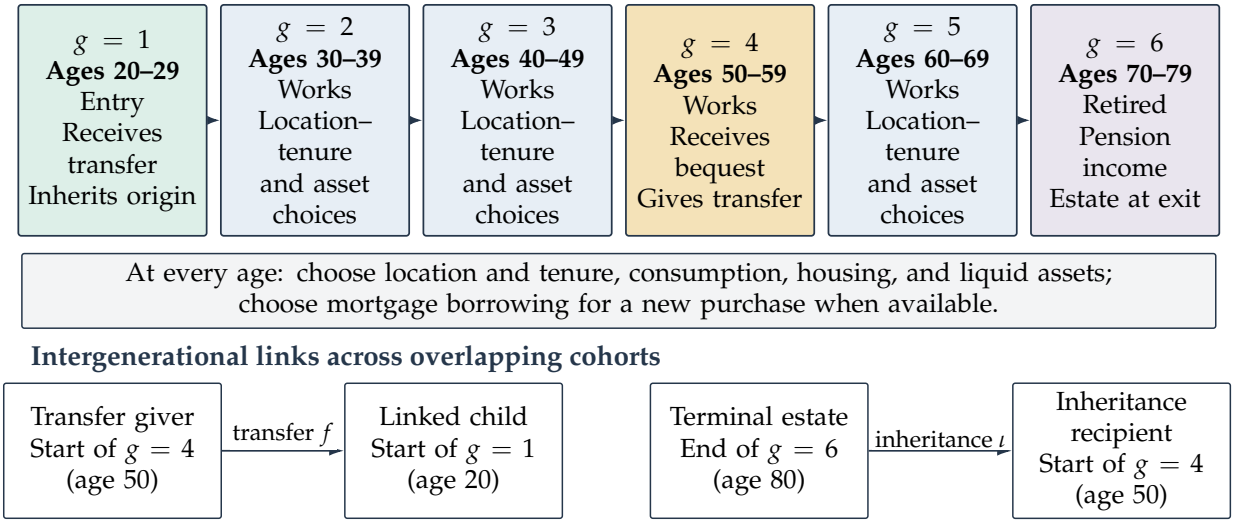


Figure 2: Six-period life cycle and intergenerational links

Notes: Each model period is ten years. A household at $g = 4$ gives transfer f to its linked $g = 1$ child. Inheritance ι is drawn from the stationary distribution of terminal estates. Estates are left at the end of $g = 6$, at the age-80 boundary, while inheritance is received at the beginning of $g = 4$, at age 50. This timing preserves the same 30-year generational gap as the inter vivos transfer link.

3.3 Housing, Mobility, and Mortgage Contracts

In each region $\ell \in \{A, B\}$, households take the purchase price $P_{\ell,t}$ and rent $R_{\ell,t}$ as given. A renter pays $R_{\ell,t}h$ for current housing services. A buyer pays $P_{\ell,t}h$ and carries the house and any mortgage into the next age. An incumbent owner may instead keep the current house, leaving its size fixed and ruling out a new mortgage.

Let ℓ_-, h_-, and m denote the location, size, and outstanding mortgage balance of a house owned at the beginning of the period. Housing depreciates at the model-period rate δ_h. If the household sells the house, the resources released by the sale are its home equity,

$$\mathcal{E}(\ell_-, h_-, m) = (1 - \delta_h)P_{\ell_-,t}h_- - m. \tag{5}$$

Sale equity can finance renting or another purchase. A keeper does not realize this equity and instead makes the scheduled mortgage payment. The feasible state space excludes negative home equity.

To capture institutional barriers such as China's *Hukou* system, a household moving from Region B to either rA or oA pays the age-specific entry cost $\kappa^{\text{entry}}_{A,g}$; an age-1 entrant's previous location is its parental origin. Choosing rA also incurs the recurring friction

$\kappa^{\text{rent}}_{rA,g}$. This wedge captures costs associated with renters' weaker access to housing-linked schools, health care, and other local services, which often depends on residence permits and registered leases (Zhang and Chen, 2018, General Office of the State Council of China, 2016, General Office of the Guangzhou Municipal Government, 2017). The combined cost is

$$\kappa^g(j;\ell_-) = \kappa^{\text{entry}}_{A,g}\mathbb{1}\{\ell_- = B,\ j \in \{rA, oA\}\} + \kappa^{\text{rent}}_{rA,g}\mathbb{1}\{j = rA\}. \tag{6}$$

The entry cost is paid whenever a household moves from B to A, including re-entry after a previous departure; the rental friction is paid in every period spent renting in A. Both frictions enter the budget as resource costs. A continuing homeowner in Region A pays neither.

Households can save in a liquid asset x and earn the gross return R_s; $x' \geq 0$ rules out unsecured borrowing. A new purchase can be financed with mortgage M, subject to LTV and PTI limits:

$$0 \leq M \leq \theta_{\text{LTV}} P_{\ell,t} h, \tag{7}$$

$$\Pi_{K_g}(M) \leq \theta_{\text{PTI}} y^g_\ell(z), \tag{8}$$

where K_g is the available mortgage term and $\Pi_{K_g}(M)$ is the payment over the current model period. The LTV limit determines the minimum equity contribution, while the PTI limit ties debt service to gross labor income. Within these limits, the household budget determines whether liquid wealth, disposable income, sale equity, and net family transfers can cover the down payment, entry cost, and debt service while maintaining positive consumption and nonnegative saving.

Mortgages are fully amortizing. Let q denote the number of ten-year payment blocks remaining at the beginning of the period. A mortgage balance m requires the payment $\Pi_q(m) = \pi_q m$ and evolves according to

$$m' = \vartheta_q m, \qquad q' = \max\{q-1, 0\}, \tag{9}$$

where π_q and ϑ_q are determined by the mortgage rate and remaining maturity. A new mortgage begins with $q = K_g$ and balance M. When $q = 1$, it is repaid within the period. Selling repays the outstanding balance, whereas keeping preserves the scheduled amortization and rules out refinancing. New contracts are available only with a new purchase. Appendix A.2 derives the payment and remaining-balance coefficients.

3.4 Household Decisions over the Life Cycle

The household problem has a common location–tenure decision at every age and age-specific departures generated by transfers, inheritance, retirement, and bequests. Conditional on a discrete alternative, households optimize consumption, housing, liquid saving, and, for a new purchase, mortgage debt. In a stationary equilibrium, prices and distributions are time invariant; calendar-time subscripts are suppressed in the household problem below.

At ages 20–29, the household state is

$$s^1 = (z, f, \ell^{\text{org}}), \tag{10}$$

where z is lifetime productivity, f is the parental transfer received upon entry, and ℓ^{org} is the parent's beginning-of-period region. The entrant begins without liquid assets, housing, or mortgage debt. For ages 30–79, the state is

$$s^g = (z, x, \ell_-, \pi_-, h_-, m, q; z_c \mathbb{1}\{g = 4\}), \tag{11}$$

where x denotes liquid assets; (ℓ_-, π_-) previous location and tenure; and (h_-, m, q) the housing and mortgage position of an incumbent owner. At ages 50–59, the household also observes the productivity z_c of its linked adult child entering ages 20–29.

The feasible location–tenure alternatives are

$$\mathcal{J}_g(s^g) = \begin{cases} \{rB, oB, rA, oA\}, & g = 1 \text{ or } \pi_- = r, \\ \{rB, oB, rA, oA, k\ell_-\}, & g \geq 2 \text{ and } \pi_- = o, \end{cases} \tag{12}$$

where $k\ell_-$ denotes keeping the current house. Entrants and renters choose among the four location–tenure alternatives; incumbent owners can also keep.

Let $v_j^g(s^g)$ denote the value conditional on alternative j after optimizing its controls. After observing the location–tenure taste shocks $\boldsymbol{\nu}$, the household chooses its preferred alternative and has value

$$V^g(s^g, \boldsymbol{\nu}) = \max_{j \in \mathcal{J}_g(s^g)} \left\{ v_j^g(s^g) + \nu_j \right\}. \tag{13}$$

Before the shocks are realized, its expected value is, up to an additive constant,

$$\overline{V}^g(s^g) = \sigma_\nu \log \left[\sum_{j \in \mathcal{J}_g(s^g)} \exp\left(\frac{v_j^g(s^g)}{\sigma_\nu} \right) \right], \tag{14}$$

and the probability of choosing alternative j is

$$\overset{g}{\Pr}(j \mid s^g) = \frac{\exp\big(v_j^g(s^g)/\sigma_\nu\big)}{\sum_{k \in \mathcal{J}_g(s^g)} \exp(v_k^g(s^g)/\sigma_\nu)}. \tag{15}$$

Disposable household income in region ℓ is

$$Y_\ell^{g,\mathrm{disp}}(z) = \begin{cases} (1-\tau_l) y_\ell^g(z) + T, & g \le 5, \\ y^{\mathrm{pen}}(z) + T, & g = 6. \end{cases} \tag{16}$$

Here $y_\ell^g(z) = w\,\mathrm{TFP}_\ell G_g z$ is gross labor income, $y^{\mathrm{pen}}(z) = \rho_{\mathrm{pen}} w G_4 z$ is pension income, and T is the universal government transfer. Let $f_g^{\mathrm{in}} = f$ at ages 20–29 and zero otherwise, and let $f_g^{\mathrm{out}} = f$ at ages 50–59 and zero otherwise. If an incumbent owner sells, $\mathcal{E}(s^g)$ denotes the home equity in (5); it is zero for entrants and incumbent renters.

For a renter choosing region $\ell \in \{A, B\}$, the budget constraint is

$$\begin{gathered} c + x' + f_g^{\mathrm{out}} + R_\ell h + \kappa^g(r\ell; \ell_-) = Y_\ell^{g,\mathrm{disp}}(z) + R_s x + f_g^{\mathrm{in}} + \mathcal{E}(s^g), \\ x' \ge 0, \qquad h \ge \underline{h}. \end{gathered} \tag{17}$$

Here $\underline{h}$ denotes the minimum dwelling size. Because renters cannot originate a mortgage and must choose nonnegative liquid assets, they cannot borrow.

For a household purchasing a new house in region ℓ, the budget is

$$\begin{gathered} c + x' + f_g^{\mathrm{out}} + P_\ell h - M + \Pi_{K_g}(M) + \kappa^g(o\ell; \ell_-) = Y_\ell^{g,\mathrm{disp}}(z) + R_s x + f_g^{\mathrm{in}} + \mathcal{E}(s^g), \\ x' \ge 0, \qquad h \ge \underline{h}. \end{gathered} \tag{18}$$

The new mortgage is subject to

$$\begin{aligned} 0 \le M &\le \theta_{\mathrm{LTV}} P_\ell h, \\ \Pi_{K_g}(M) &\le \theta_{\mathrm{PTI}} y_\ell^g(z). \end{aligned} \tag{19}$$

The first restriction imposes the LTV ceiling; the second limits debt service relative to gross labor income. Family transfers expand the resources available in the purchase budget but are excluded from the gross-income base used to calculate the PTI borrowing limit.

An incumbent owner that keeps its current house faces

$$c + x' + f_g^{\text{out}} + \Pi_q(m) = Y_{\ell_-}^{g,\text{disp}}(z) + R_s x + f_g^{\text{in}}, \qquad x' \geq 0. \tag{20}$$

The keeper neither realizes sale equity nor pays rent, a purchase price, or a mobility cost; it retains its housing size, and its mortgage follows the scheduled transition in Appendix A.2.

For ages 20–49 and 60–69, conditional values have the common form

$$v_j^g(s^g) = \max_{\boldsymbol{a}_j \in \Gamma_j^g(s^g)} \left\{ u^g(c,h,j) + \beta \mathbb{E}\left[\overline{V}^{g+1}(s^{g+1}) \right] \right\}. \tag{21}$$

For renters, new buyers, and keepers, respectively, $\boldsymbol{a}_j$ is (c, h, x'), (c, h, x', M), and (c, x'); Γ_j^g imposes the corresponding budget, financing constraints, and state transition. Although equation (21) has the same recursive form at these ages, their feasible sets differ. At ages 20–29, f is a predetermined transfer received from the parent, $x = \mathcal{E} = 0$, and ℓ^{org} determines whether the Region A entry cost applies. No family transfer enters the budget at ages 30–49 or 60–69. At ages 40–49, the expectation also integrates over inheritance received upon entry into ages 50–59.

Upon entering ages 50–59, the household receives an inheritance ι, which augments its liquid-asset state:

$$x_4 = x_3' + \frac{\iota}{R_s}. \tag{22}$$

Thus, inheritance raises current resources one-for-one.

The parent observes the productivity z_c of its linked child entering ages 20–29. The child's origin is the parent's beginning-of-period region ℓ_- and is unaffected by the parent's current location choice. The altruistic parent chooses a one-time monetary transfer $f \geq 0$ jointly with its own location, housing, and saving decisions. It anticipates the child's optimal response given z_c, f, and ℓ_-, averaging over the child's idiosyncratic location–tenure taste shocks. Conditional on its own alternative j, the parent chooses the action-specific controls $\boldsymbol{a}_j$ and transfer f:

$$v_j^4(s^4) = \max_{(\boldsymbol{a}_j, f) \in \Gamma_j^4(s^4)} \left\{ u^4(c,h,j) + \beta \mathbb{E}\left[\overline{V}^5(s^5) \right] + \alpha_p \overline{V}^1(z_c, f, \ell_-) \right\}, \qquad f \geq 0. \tag{23}$$

The altruistic term $\alpha_p \overline{V}^1(z_c, f, \ell_-)$ places weight α_p on the child's optimized expected lifetime utility at the chosen transfer. Parental giving expands the child's budget while

reducing the resources available for the parent's own consumption, housing, and saving.

At ages 70–79, pension income replaces labor income and is independent of retirement location. New mortgage borrowing is unavailable, so a purchase must be financed from liquid resources and realized home equity. The continuation value is replaced by the warm-glow utility in (4), evaluated at the terminal estate

$$e = x' + \mathbb{1}\{\pi' = o\}\left[(1-\delta_h)P_{\ell'}h' - m'\right]. \tag{24}$$

Terminal estates generate the inheritance distribution received at ages 50–59. Appendix A.1 records the state transitions and feasibility restrictions.

Renting provides access to Region A's labor market without a down payment, whereas ownership requires upfront financing and builds housing equity. Family transfers relax the child's financing constraint and can change both location and tenure, at a cost to the parent's consumption, housing, and saving.

3.5 Firms

3.5.1 Consumption-Good Producer

The homogeneous consumption good is freely traded across regions and is the numeraire. In region $\ell \in \{A, B\}$, a representative competitive firm uses a linear technology. Let $\mu^g_{\ell,t}$ denote the measure of age-g workers residing in region ℓ. These measures are expressed relative to the total population: each age group has mass $1/6$ across the two regions, so age weights are already incorporated. Regional effective labor and output are

$$L_{\ell,t} = \sum_{g=1}^{5} \int G_g z \, d\mu^g_{\ell,t}(z), \qquad Y_{\ell,t} = \mathrm{TFP}_\ell L_{\ell,t}. \tag{25}$$

Only ages $g = 1, \ldots, 5$ supply labor. Linearity, competition, and positive production imply a normalized wage of one per productivity-adjusted efficiency unit. This normalization is distinct from the monetary wage scale w in household budgets, which is fixed in the calibration.

3.5.2 Housing Supply

Residential investment in region $\ell \in \{A, B\}$ follows the reduced-form supply schedule

$$I_{\ell,t} = \bar{I}_\ell P^{\rho_\ell}_{\ell,t}, \tag{26}$$

where $I_{\ell,t}$ is newly constructed floor space, $\bar{I}_\ell$ is a region-specific supply shifter, and ρ_ℓ is the price elasticity of housing supply. Allowing both parameters to vary by region captures differences in land availability and construction constraints. The housing stock evolves as

$$H_{\ell,t} = (1 - \delta_h) H_{\ell,t-1} + I_{\ell,t}. \tag{27}$$

Owner-occupied and rental housing constitute separate market segments, but the allocation of the regional stock across them can respond to relative returns (Mabille, 2023, Greenwald and Guren, 2025). Let $s^r_{\ell,t}$ denote the rental share of the housing stock and let $(P_{\ell,0}, R_{\ell,0}, s^r_{\ell,0})$ denote its reference-equilibrium values. The rental share satisfies

$$\log\left(\frac{s^r_{\ell,t}}{1 - s^r_{\ell,t}}\right) = \log\left(\frac{s^r_{\ell,0}}{1 - s^r_{\ell,0}}\right) + \chi_\ell \left[\log\left(\frac{R_{\ell,t}}{R_{\ell,0}}\right) - \log\left(\frac{P_{\ell,t}}{P_{\ell,0}}\right)\right], \tag{28}$$

and segment supplies are

$$H^{\text{rent}}_{\ell,t} = s^r_{\ell,t} H_{\ell,t}, \qquad H^{\text{own}}_{\ell,t} = (1 - s^r_{\ell,t}) H_{\ell,t}. \tag{29}$$

An increase in rent relative to the purchase price shifts housing toward the rental segment. The parameter χ_ℓ governs this reallocation: when it is zero, the owner–rental split is fixed; positive values permit partial conversion while preserving segmentation between the two markets. Appendix A.4 gives the closed-form rental-share rule, the reference-equilibrium supply backout, and the segment-market clearing conditions.

3.6 Government

The government levies a proportional tax τ_l on the gross labor income of working-age households. Retirees do not pay labor-income tax and receive the pension $y^{\text{pen}}(z)$ defined above. Every household also receives the uniform lump-sum transfer T_t. Family transfers and inheritances are private and are neither taxed nor counted as government expenditure in the baseline.

Let $\mu^g_{\ell,t}$ denote the measure of age-g households residing in region ℓ. The government does not issue debt, and its period budget constraint is

$$\tau_l \sum_{g=1}^{5} \sum_{\ell \in \{A,B\}} \int y^g_\ell(z)\, d\mu^g_{\ell,t}(z) = \sum_{\ell \in \{A,B\}} \int y^{\text{pen}}(z)\, d\mu^6_{\ell,t}(z) + T_t \sum_{g=1}^{6} \sum_{\ell \in \{A,B\}} \int d\mu^g_{\ell,t}(z). \tag{30}$$

Thus, T_t adjusts so that labor-tax revenue equals spending on pensions and universal transfers.

3.7 Equilibrium

The model is solved as a stationary recursive equilibrium, so prices, rents, policies, and distributions are time invariant. Because production is linear and labor supply is inelastic, regional effective labor is determined by households' location choices; there is no separate labor-demand margin.

In the housing market, stationarity requires residential investment to replace depreciated housing, which implies

$$H_\ell = \frac{\bar{I}_\ell P_\ell^{\rho_\ell}}{\delta_h}. \tag{31}$$

Let $\mu^g_{o\ell}$ and $\mu^g_{r\ell}$ denote the stationary measures of owners and renters in region ℓ. The owner-occupied and rental segments clear separately:

$$\sum_{g=1}^{6} \int h \, d\mu^g_{o\ell} = (1 - s^r_\ell) H_\ell, \tag{32}$$

$$\sum_{g=1}^{6} \int h \, d\mu^g_{r\ell} = s^r_\ell H_\ell, \qquad \ell \in \{A, B\}, \tag{33}$$

where s^r_ℓ follows (28). The four prices (P_A, P_B, R_A, R_B) clear the owner and rental segments in both regions.

The gross liquid-asset return and mortgage-contract terms are exogenous. During the sample period, banks priced household deposits and mortgages within national interest-rate benchmarks and prudential rules, rather than through local housing-market clearing (People's Bank of China, 2024). The financial sector elastically absorbs household saving and supplies mortgage credit at these rates; consequently, aggregate liquid assets need not equal aggregate mortgage debt. Their difference is reported as net financial assets, but it is not an equilibrium condition. The uniform transfer T adjusts to satisfy the government budget constraint (30).

Let $\mathcal{P}$ collect the household policy functions and location–tenure choice probabilities, let μ denote the distribution over age and household states, and let $\mathcal{B}$ denote the inheritance distribution. The policies and the demographic, productivity, and intergenerational transitions described above induce a law of motion $\mathcal{T}$ for the cross-sectional distribution.

Stationarity requires

$$\mu = \mathcal{T}(\mu; \mathcal{P}, \mathcal{B}). \tag{34}$$

The inheritance distribution must also be internally consistent: the discrete distribution from which households entering age group $g = 4$ draw must coincide with the corresponding discretization of terminal net estates generated by age-6 households under μ and $\mathcal{P}$.

A stationary recursive equilibrium consists of household value and policy functions $\mathcal{P}$, a stationary distribution μ, an inheritance distribution $\mathcal{B}$, regional housing stocks and rental shares, and (P_A, P_B, R_A, R_B, T) such that households optimize, competitive production satisfies (25), all four housing-market segments clear, the government budget balances, and the cross-sectional and inheritance distributions are stationary and mutually consistent.

4 Calibration

I calibrate the stationary equilibrium using the 2017 China Household Finance Survey (CHFS) for earnings, location, tenure, and housing-market moments, and the 2018 China Household Income Project (CHIP) for parental housing assistance. The sample is restricted to urban households, whose housing and labor markets correspond to the model. Under the city-tier classification used in this paper, Region A comprises 18 Tier 1 and Tier 2 cities; Region B contains the remaining urban cities. CHFS has a larger household sample and broader city coverage than CHIP, which represents only 13 of these 18 cities, omitting Shanghai, Hangzhou, Tianjin, Xi'an, and Ningbo. Region A's geographic coverage therefore differs from that in Section 2.

Table 2 distinguishes internally calibrated parameters, estimates, and external assignments. Monetary values are in units of 10,000 RMB, with housing prices measured per square meter. Appendix B documents the samples, moment construction, and parameter sources; Table B.5 there collects the full age-dependent schedules.

4.1 Family Transfers and Location–Tenure Sorting

I jointly calibrate parental altruism α_p, the taste-shock scale σ_ν, and 18 age-specific location–tenure constants $\{\xi_j^g\}$. The altruism target is parental housing assistance among young homeowners: 275 of the 438 current homeowners in the CHIP sample of household heads aged 20–29 report assistance, giving a target of 62.79 percent. The calibrated $\alpha_p = 0.4908$ yields 62.37 percent. Appendix B.6 documents the survey question and its

mapping to the model's monetary transfer.

Normalizing $\xi^g_{rB} = 0$, the other three constants fit the three independent location–tenure shares in each age group. They absorb residual differences in nonpecuniary attractiveness after accounting for earnings, housing costs, mobility costs, and financing constraints. I estimate the age-efficiency profile and permanent-productivity distribution from CHFS earnings and obtain a regional total factor productivity (TFP) ratio of 1.37. Conditional on this productivity premium, $\sigma_\nu = 0.0146$ targets a working-age A/B average gross-income ratio of 2.0, disciplining endogenous talent sorting. This is an equilibrium sorting target rather than the raw survey income ratio. Appendix B.3 documents income cleaning, the earnings regressions, and construction of the talent distribution, and reports the raw and adjusted regional income ratios.

4.2 Rental Access and Mobility Costs

The recurring Region A rental friction targets the direction and approximate size of the observed productivity gap between young A renters and B owners. The productivity gap is 9.0 percent among CHFS household heads and 11.4 percent in the benchmark; removing the friction reduces the model gap to 4.8 percent. Appendix B.4 compares this productivity gap across samples of young household heads and all young workers. Appendix D.5 provides sensitivity tests of the transfer effects on young households' residence and homeownership in Region A by removing or doubling the rental friction.

The friction represents recurring resource costs of renting in A, including unequal access to housing-linked local public services (Zhang and Chen, 2018, General Office of the State Council of China, 2016, General Office of the Guangzhou Municipal Government, 2017). Entry costs are assigned externally using migration-cost evidence (Morten and Oliveira, 2016, Kennan and Walker, 2011): CHFS does not separately identify them from location preferences. The full age profiles and their supporting evidence appear in Appendix B.4.

4.3 Housing-Market Adjustment and Model Fit

Housing-supply elasticities govern the construction response to purchase prices. I assign $\rho_A = 0.65$ and $\rho_B = 0.95$, guided by Chinese estimates for more and less regulated cities (Huang, 2022); the lower value in A captures tighter land and regulatory constraints. Comparable regional estimates of owner–rental conversion are unavailable, so I set $\chi_A = \chi_B = 1$ as a unit-elastic benchmark for rental-to-owner floor-space odds. This permits partial reallocation between tenures. Appendix D.4 examines alternative values of both sets of elasticities.

Table 2: Calibration: Main Parameters

Parameter	Description	Value	Discipline or source
Panel A. Internally calibrated			
α_p	Parental altruism	0.4908	Owner receipt: $275/438 = 62.79\%$; model: 62.37%
σ_ν	Taste-shock scale	0.0146	Working-age A/B income-ratio target: 2.0
$\{\xi_j^g\}$	Location–tenure constants	Appendix Table B.5, Panel B	Three independent choice shares per age
$\kappa_{rA,g}^{\text{rent}}$	A rental friction	2–3, by age	Young rA–oB productivity-gap target: 9.0%
$\Omega_{A,0}, \Omega_{B,0}$	Owner floor-space shares	0.551, 0.788	CHFS purchase prices and rents
$\bar{I}_A, \bar{I}_B$	Construction shifters	1.381, 9.686	CHFS purchase prices and rents
T	Universal transfer	0.4293	Balanced government budget
Panel B. Estimated from data and normalizations			
$\{G_g\}$	Age-efficiency profile	Appendix Table B.5, Panel A	CHFS earnings regression
$\text{TFP}_A, \text{TFP}_B$	Regional productivity	1.37, 1	CHFS earnings regression
w^{ann}	Annual earnings scale	5.000	Mean $z = 1$; RMB 50,000
Panel C. Externally assigned			
β	Ten-year discount factor	0.5987	Annual factor: 0.95
γ	Risk aversion	2.000	Mabille (2023)
ϵ	CES curvature	-2.000	Substitution elasticity: $1/3$
α_h	Housing utility weight	0.500	Equal CES weights
ω_b	Warm-glow weight	0.100	De Nardi (2004)
$\kappa_{A,g}^{\text{entry}}$	A entry cost	4–9, by age	Morten and Oliveira (2016)
ρ_A, ρ_B	Housing-supply elasticities	0.65, 0.95	Huang (2022)
χ_A, χ_B	Owner–rental conversion	1.00, 1.00	Unit-elastic rental-to-owner space odds
δ_h	Ten-year depreciation	0.140	Annual rate: 1.5%; Kaplan et al. (2020)
r_s^{ann}	Annual deposit rate	1.75%	Bank of China, 2015
r_m^{ann}	Annual mortgage rate	5.01%	PBC, 2017:Q3
θ_{LTV}	LTV ceiling	0.70	Minimum down payment: 30%
θ_{PTI}	PTI ceiling	0.50	Fang et al. (2016)
$\{K_g\}$	Mortgage term	0–3 ten-year blocks	30-year maximum; Bank of China (2008)
ρ_{pen}	Pension replacement	0.50	Fang and Feng (2020)
τ_l	Labor-income tax	0.10	Urban average tax rate

Notes: Owner receipt conditions on current homeownership in either region among CHIP household heads aged 20–29. The rental-friction target is matched approximately. $\xi_{rB}^g = 0$ at every age. Full age schedules appear in Appendix Table B.5; additional parameters and sources appear in Appendix Table B.4 and Appendix B.7. Monetary values are in 10,000 RMB.

Conditional on these elasticities, I back out the reference owner-space shares $\Omega_{\ell,0}$ and construction shifters $\bar{I}_\ell$ so that supply equals model floor-space demand at the CHFS purchase prices and rents. Appendix B.5 details this backout. The universal government transfer balances the fiscal budget, and the inheritance distribution matches terminal estates in the stationary equilibrium. The remaining assigned preference, financial, and institutional parameters are summarized in Table 2 and documented in Appendix B.7. Table 3 reports the targeted choice shares, prices, rents, income ratio, productivity gap, and homeowner-receipt rate alongside their model counterparts.

Table 3: Targeted Moments and Model Fit

	rB		oB		rA		oA	
Age group	Target	Model	Target	Model	Target	Model	Target	Model
Panel A. Location–tenure shares (percent)								
20–29	26.00	25.14	39.00	39.75	22.05	21.47	12.95	13.63
30–39	9.75	9.47	55.25	55.32	14.00	13.70	21.00	21.50
40–49	10.50	10.04	59.50	59.55	9.00	8.81	21.00	21.59
50–59	10.50	10.45	59.50	59.66	9.00	9.05	21.00	20.84
60–69	9.75	9.76	55.25	55.47	10.50	10.51	24.50	24.26
70–79	9.75	9.83	55.25	55.69	10.50	10.42	24.50	24.07

Panel B. Prices, spatial sorting, and parental assistance		
Moment	Target	Model
Region A purchase price	1.837	1.837
Region B purchase price	0.475	0.475
Region A ten-year rent	0.311	0.311
Region B ten-year rent	0.158	0.158
Working-age average gross-income ratio, A/B	2.000	1.968
Mean-productivity gap, young rA vs. oB (percent)	9.0	11.4
Parental assistance among young homeowners (percent)	62.79	62.37

Notes: The four alternatives are renting in Region B (rB), owning in Region B (oB), renting in Region A (rA), and owning in Region A (oA). Purchase prices and ten-year rents are in units of 10,000 RMB per square meter. The income ratio is computed among working-age households. The productivity-gap target is $100(E[z \mid rA]/E[z \mid oB] - 1)$ for household heads aged 20–29; the rental-friction schedule approximately matches this moment. Parental-assistance receipt is conditional on ownership in either region among households headed by ages 20–29 in CHIP 2018: 275 recipients among 438 current homeowners, or 62.79 percent. All moments reported in the table enter the calibration.

5 Steady-State Results

This section describes the baseline allocation of households and parental support, then examines how family resources shape location, tenure, and housing finance.

5.1 Baseline Allocation and Transfer Receipt

Homeownership rises sharply early in adult life, from 53.4 percent at ages 20–29 to 76.8 percent at ages 30–39, while Region A residence remains around 35 percent. Ownership remains high thereafter, with comparatively modest changes in regional residence (Table 3 and Online Appendix Figure S1).

Figure 3 shows positive spatial sorting among young households. Residence in A becomes more likely as talent rises, but renting there is most prevalent at intermediate talent, while ownership increases steadily with talent. Mean talent is 1.012 among young A renters, compared with 0.908 among B owners and 1.391 among A owners (Online Appendix Table S4). Renting allows households to live and work in A without financing a home purchase.

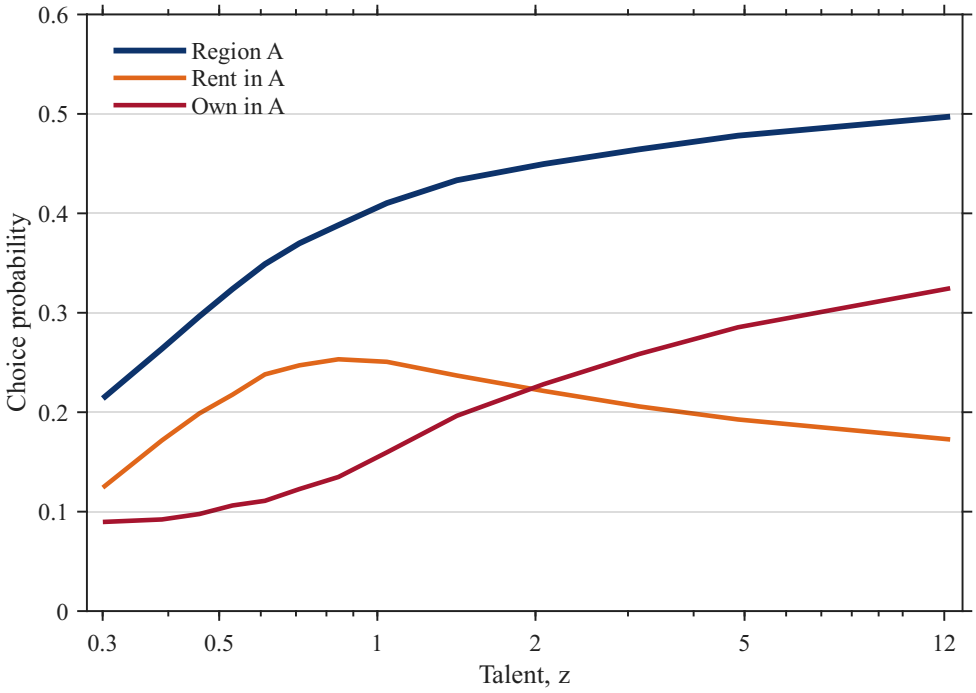


Figure 3: Region A Location Choices by Talent, Ages 20–29

Notes: Each line reports the model-implied choice probability conditional on talent among households aged 20–29. Region A includes both renters and owners in Region A. The talent axis is shown on a logarithmic scale.

In the baseline, 58.0 percent of all households aged 20–29 receive a positive transfer, and mean support across all young households, including nonrecipients, is approximately RMB 217,000. Figure 4 shows that lower-talent children receive more support, while higher-talent parents give more. Children's financing needs and parents' capacity to give jointly shape this allocation: the highest-talent 1 percent of parents supply 13.7 percent of total transfers.

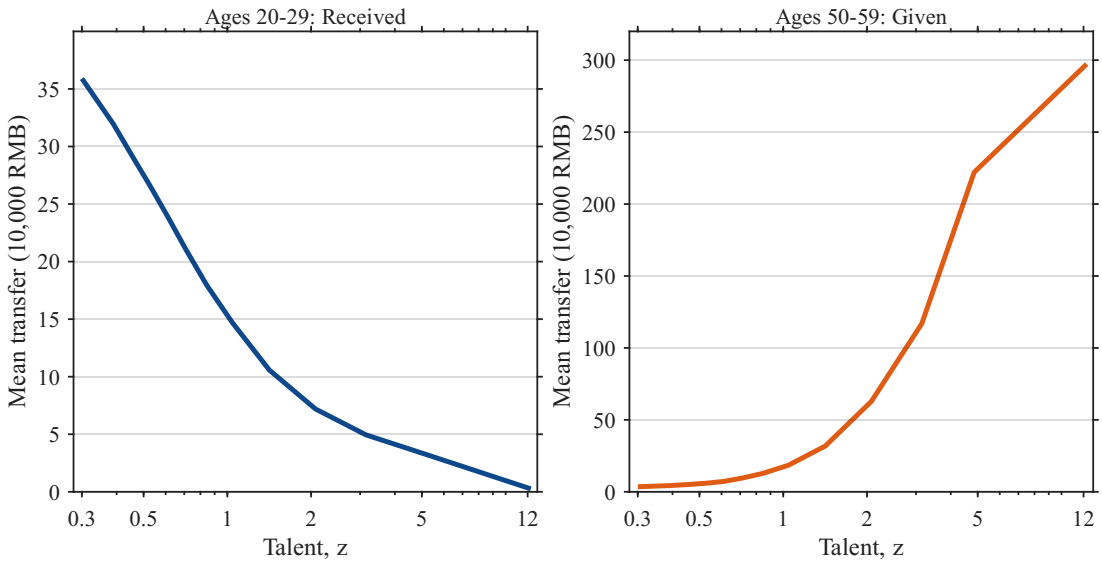


Figure 4: Intergenerational Transfers by Recipient and Giver Talent

Notes: The left panel reports mean transfers received by households aged 20–29 conditional on the recipient's talent. The right panel reports mean transfers given by households aged 50–59 conditional on the giver's talent. Means include zero transfers. Monetary values are expressed in units of 10,000 RMB, and the talent axes are shown on logarithmic scales.

5.2 Family Resources and Location–Tenure Choices

Family background matters most for children with low earning capacity. Conditional on low child talent, children of wealthier parents are substantially more likely to reside and own in A; these differences narrow with talent and become negligible at high talent levels. Section 6 examines how allowing transfers changes these disparities.

Figure 5 traces location–tenure choices as parental support varies for households at the 10th, 50th, and 90th talent percentiles. Holding talent and the economic environment fixed separates these financing responses from differences in who receives support.

Ownership in A rises with support at all three talent levels, most strongly for low- and intermediate-talent households. Large transfers narrow their ownership gaps with high-talent households. Rental access limits the associated residence response: all three groups can rent in A without parental support, and at high talent the increase in A ownership is almost entirely offset by lower A renting, leaving residence nearly unchanged.

Small transfers instead make cheaper ownership in B more attractive. Raising support from zero to RMB 20,000 increases low-talent B ownership from 29.1 to 39.7 percent, while A ownership remains close to zero and A residence falls. At intermediate talent, support up to RMB 200,000 increases ownership in both regions, but the decline in A renting exceeds the increase in A ownership. Beyond RMB 200,000, low-talent A residence rises as both owning and renting there become more likely. The location response therefore

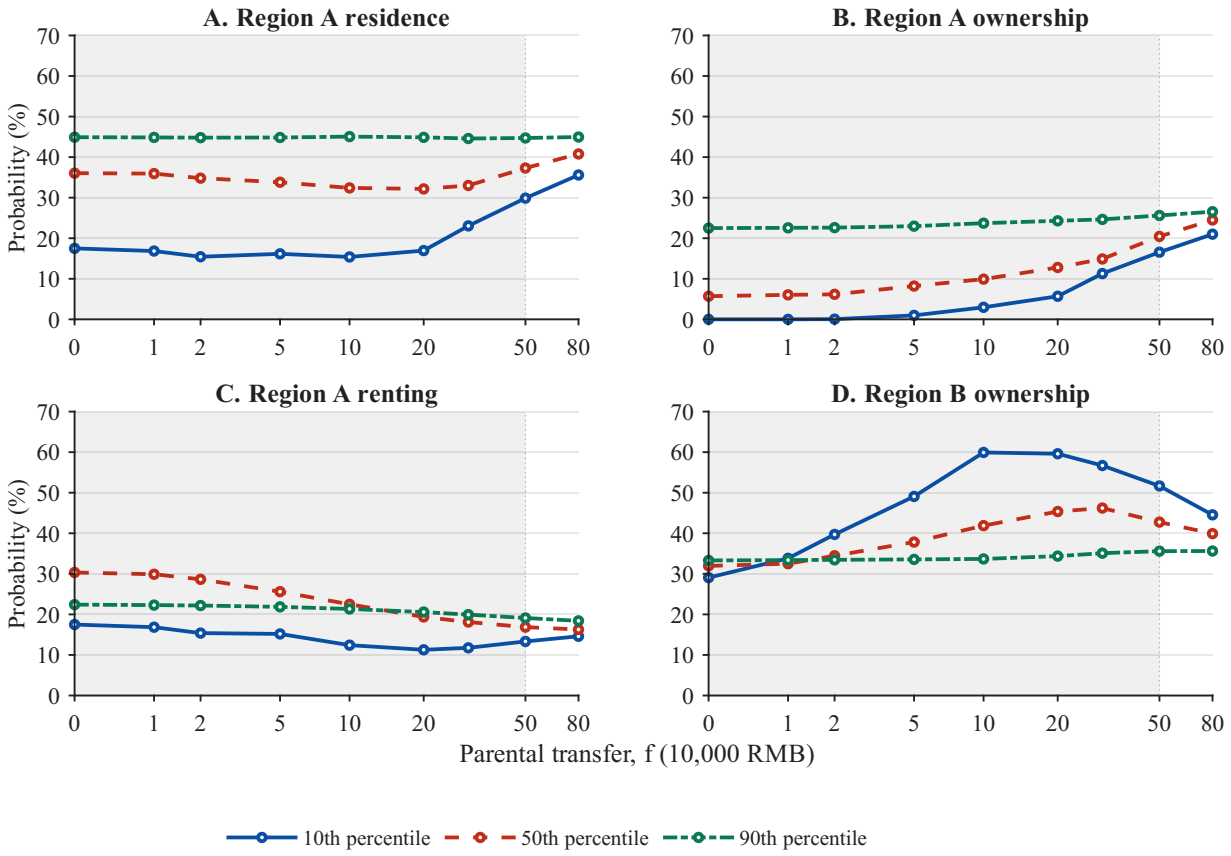


Figure 5: Parental Transfers and Location–Tenure Choices, Ages 20–29

Notes: Panels A–C show residence, ownership, and renting in Region A; Panel D shows ownership in Region B. All panels use the same 10th, 50th, and 90th talent percentiles and probability scale. Ownership and renting are joint location–tenure probabilities; Panels B and C sum to Panel A. Transfers are in 10,000 RMB on a $\log(1 + f)$ scale; probabilities are in percent. Shading covers transfers up to RMB 500,000 and accounts for 91.3 percent of young households, including nonrecipients. These three profiles are slices of the full talent–transfer grid (Online Appendix Figure S5).

depends on talent and transfer size, even as support raises ownership in A.

The recipient–nonrecipient comparison parallels Section 2. Among young households residing in A, ownership is 41.4 percent among recipients and 36.1 percent among nonrecipients. The share residing in A is lower among recipients, at 31.5 percent compared with 40.0 percent among nonrecipients. Appendix Table B.6 reports the full comparison with CHIP.

Recipients' lower Region A residence share compares households that differ in earning capacity and parental resources. Assistance can make a household more likely to live in A while leaving it less likely to do so than a nonrecipient with higher earning capacity. Section 6 therefore compares the residence share of all young households across economies with and without inter vivos transfers, allowing prices, rents, and household choices to adjust. This comparison measures whether allowing transfers increases aggre-

gate residence in A, rather than whether recipients reside there more often than nonrecipients.

5.3 Housing Finance

Buying in A entails a larger upfront financing burden: down payments among young owners amount to 50.1 percent of ten-year gross income, compared with 42.4 percent in B. Table 4 compares young owners' purchase values, financing, parental support, and equity. Mean transfers to A owners equal 85.1 percent of mean down payments (Online Appendix Table S1). This ratio measures support relative to financing needs; it does not trace how fungible gifts are spent.

Table 4: Housing Finance among Young Homeowners

Moment	Owners in A	Owners in B
Mean purchase value	70.35	23.59
Down payment / ten-year gross income (percent)	50.1	42.4
Origination LTV (percent)	44.1	32.9
Mean parental transfer	34.34	22.75
Transfer / down payment (percent)	85.1	140.0
Mean end-of-age-1 home equity	45.92	17.62

Notes: Entries concern homeowners aged 20–29 and use stationary state mass and choice-probability weights. Monetary values are in 10,000 RMB; mean transfers include zero receipts. Affordability ratios use weighted group totals and ten-year gross income. Origination debt is recovered from the post-amortization balance; equity is measured at the end of age 1. Transfer-to-down-payment ratios do not trace the use of individual gifts.

Mortgage leverage declines with age, while mean home equity rises through ages 50–59 before declining (Online Appendix Table S2). Section 6 compares these balance sheets with the no-transfer equilibrium.

6 The Role of Family Transfers

To quantify the equilibrium role of inter vivos family transfers, I compare the baseline economy B with a counterfactual stationary equilibrium B^0 in which the age-4-to-age-1 monetary transfer is constrained to zero in every state. Parents remain altruistic, but they can no longer implement this concern through an inter vivos payment. The equilibrium is re-solved, allowing housing prices, rents, household choices, and the distribution of inheritances to adjust. Unless otherwise stated, effects are reported as the difference between the transfer economy and the no-transfer economy, $B - B^0$. Percentage changes use the no-transfer equilibrium as the reference.

6.1 Spatial Access versus Ownership Access

Family transfers expand young households' ownership in the productive region substantially more than their residence there. Among households aged 20–29, transfers raise the share living in Region A by 1.18 percentage points, or 3.5 percent relative to the no-transfer equilibrium. The share owning a home in Region A increases by 4.30 percentage points, or 46.0 percent. Ownership also expands in Region B, bringing the increase in homeownership among young households to 9.21 percentage points (Online Appendix Table S3).

Table 5 separates the increase in Region A ownership into a spatial component, associated with the change in Region A residence, and a tenure component, associated with the change in ownership conditional on residence. For young households, the tenure component accounts for 3.90 percentage points, or 90.9 percent of the ownership response, while the spatial component accounts for 0.39 points. Appendix C.1 presents the decomposition. The larger tenure adjustment is also reflected in the share of young households renting in Region A, which is 3.12 percentage points lower in the transfer economy. These results give an equilibrium answer to the distinction raised in Section 2: family finance matters principally for owning within the productive region, while rental housing allows many households to access that region without purchasing.

Table 5: Residence–Ownership Decomposition of the Family-Transfer Effect

Group	$\Delta \Pr(A)$	$\Delta \Pr(\text{own} \mid A)$	$\Delta \Pr(oA)$	Spatial component	Tenure component	Tenure share
Age 20–29	1.18	11.31	4.30	0.39	3.90	90.9%
Working ages	1.44	4.94	2.45	0.85	1.60	65.2%
All households	1.24	3.52	1.91	0.76	1.15	60.3%

Notes: Changes are $B - B^0$. All changes and components are in percentage points. Appendix C.1 defines the components and tenure share. Working ages comprise ages 20–69. All statistics use stationary population weights.

Transfers do not uniformly encourage residence in Region A. The conditional choices in Section 5.2 show that modest support makes ownership in lower-cost Region B more attractive relative to renting in A. Larger transfers instead help finance purchases in A and increase the probability of residing there. Family support therefore promotes ownership through choices that have different implications for location. In the full equilibrium, where parents choose transfers and housing prices and rents adjust, ownership rises in both regions, but the net increase in Region A residence is comparatively small.

Figure 6 shows that the spatial response is concentrated among young households with low talent. At the lowest talent node, transfers raise Region A residence by 5.84

percentage points and Region A ownership by 8.96 points. The residence effect becomes small as talent increases, while the ownership response remains appreciable over a broader range of lower- and middle-talent households. At high talent levels, both responses are small.

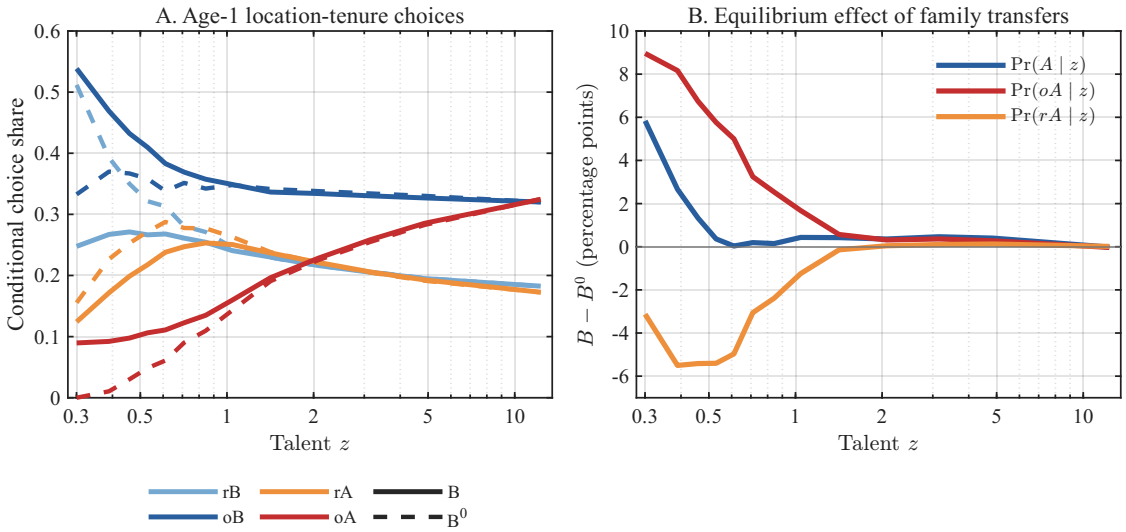


Figure 6: Family Transfers and Age-1 Location–Tenure Choices

Notes: Panel A reports age-1 choice probabilities conditional on talent. Colors identify renting and owning in Regions B and A; solid lines denote the baseline equilibrium B, and dashed lines denote the no-transfer equilibrium B^0. Panel B reports $B - B^0$ in percentage points, where Region A residence includes both rA and oA. Each curve averages over the remaining age-1 states using the stationary distribution of the corresponding equilibrium. The talent axis is logarithmic.

The responses at later ages are more heterogeneous. Higher Region A ownership among lower-talent households persists at ages 30–39 and 40–49 (Online Appendix Figure S6). From ages 50–59 onward, however, the direction of the ownership response differs more sharply across talent. Lower-talent households continue to have higher Region A ownership and lower Region B ownership in the transfer economy. Among middle- and higher-talent households, Region A ownership is instead lower, accompanied by a partial offsetting increase in ownership in the less expensive Region B. Homeownership in either region also falls for these households, particularly at ages 60–69 and 70–79.

Households can receive parental support when young and later decide how much to give to their own children. Lower-talent households receive more assistance when young and give relatively little later in life. Middle- and higher-talent households receive less support at entry but devote more resources to their own children at ages 50–59. The financing advantage associated with receiving transfers therefore coexists with a resource cost when households become donors.

These age-specific reversals do not overturn the aggregate result. When all age groups are considered together, transfers still increase Region A residence and ownership, with tenure changes accounting for most of the ownership response (Table 5).

6.2 Ownership Selection and the Spatial Allocation of Labor

Family transfers weaken the dependence of Region A ownership on young households' own earning capacity. Panel A of Table 6 shows that, among households aged 20–29 in the bottom 80 percent of the talent distribution, the probability of owning in Region A rises from 6.16 percent without transfers to 11.42 percent with transfers. The corresponding probability for the top 20 percent changes only slightly, from 22.02 to 22.45 percent. Weighting these responses by the groups' population shares, households in the bottom 80 percent account for 98.0 percent of the net increase in young Region A ownership. The expansion is therefore concentrated among households whose own earnings provide less capacity to finance a purchase.

This expansion changes the composition of young Region A homeowners. The bottom 80 percent of the talent distribution account for 67.05 percent of these owners in B, compared with 52.80 percent in B^0. Mean annual gross labor income among young A owners is consequently 22.1 percent lower in the transfer economy, at approximately RMB 80,500 rather than RMB 103,400. Parental resources enable households with lower own earnings to purchase in Region A even after housing prices adjust.

The reduction in mean talent among Region A owners persists at later ages but becomes smaller (Online Appendix Table S4). Changes in homeowner composition do not, by themselves, imply changes in aggregate output. Workers of the same age and talent supply the same effective labor in Region A whether they rent or own. The output effect therefore depends on the number and age–talent composition of all workers residing in A, including both renters and owners. Panel B of Table 6 reports an accounting decomposition of the change in Region A's effective labor into a population component, reflecting changes in its working-age population (ages 20–69), and a composition component, reflecting changes in mean age-adjusted productivity. Appendix C.1 provides the construction.

Transfers raise aggregate output by 0.23 percent through a net reallocation of effective labor toward Region A. The gain is limited by the modest residence response and lower average productivity among Region A workers: in the accounting decomposition, the composition change offsets 61.4 percent of the population contribution to A's effective labor.

Table 6: Ownership Selection, Effective Labor, and Aggregate Output

Outcome	B: transfers	B^0: no transfers	Transfer effect ($B - B^0$)
Panel A. Young-household ownership and homeowner composition			
Region A ownership among bottom-80 households	11.42%	6.16%	+5.26 pp
Region A ownership among top-20 households	22.45%	22.02%	+0.43 pp
Bottom-80 share among Region A owners	67.05%	52.80%	+14.25 pp
Mean annual labor income among Region A owners (RMB)	80,468	103,360	−22.1%
Panel B. Working-age effective labor and output			
Working-age mass in A	0.2756	0.2636	+0.0120
Mean $G_g z$ in A	1.1027	1.1333	−0.0306
Effective labor in A	0.3039	0.2988	+0.0052
Residence-mass component	—	—	+0.0134
Composition component	—	—	−0.0082
Effective labor in B	0.4282	0.4334	−0.0052
Output in A	0.4164	0.4093	+0.0071
Output in B	0.4282	0.4334	−0.0052
Aggregate output	0.8446	0.8426	+0.0019 (+0.23%)

Notes: Panel A concerns ages 20–29; Panel B concerns working ages 20–69. Effects are $B - B^0$, with percentage changes relative to B^0; differences are calculated before rounding. In Panel A, ownership rates use the respective talent group's population as the denominator, whereas the bottom-80 share uses all young Region A owners. Income is annual gross labor income, excluding family and government transfers. Population-weighted bottom-80 ownership changes account for 98.0 percent of the net increase in young Region A ownership. In Panel B, working-age mass is relative to the full stationary population. Since working ages comprise 5/6 of that population, the 1.20-percentage-point mass increase corresponds to the 1.44-point residence-share increase among working-age households in Table 5. Effective labor and its components are in efficiency units; regional output equals regional TFP times effective labor.

6.3 Housing Markets and Life-Cycle Balance Sheets

Compared with the no-transfer equilibrium, parental transfers increase young households' homeownership while raising purchase prices in both regions. The rise in purchase costs does not offset the ownership gains from family finance. Purchase prices are 4.55 percent higher in Region A and 8.98 percent higher in Region B (Table 7). Rent differences are much smaller. With transfers, rent is 0.49 percent lower in A and 0.54 percent higher in B.

Higher purchase prices support greater residential investment through the regional supply schedules. Housing stocks expand, while lower rent-to-price ratios shift their allocation toward owner use in both regions. These adjustments extend the effects beyond transfer recipients: higher purchase prices also raise acquisition costs for prospective buyers without parental assistance.

Young buyers in Region A rely less on mortgage financing in the transfer economy. Their mortgages cover an average of 44.1 percent of the purchase price, compared with 68.8 percent in the no-transfer equilibrium (Online Appendix Table S9). Together with the

Table 7: Housing-Market and Age-1 Balance-Sheet Effects of Family Transfers

Outcome	B: transfers	B^0: no transfers	Transfer effect ($B - B^0$)
Panel A. Housing markets and fiscal balance			
P_A	1.837	1.757	$+4.55\%$
P_B	0.475	0.436	$+8.98\%$
R_A	0.311	0.312	-0.49%
R_B	0.158	0.157	$+0.54\%$
Rental share of housing stock in A	44.92%	46.14%	-1.22 pp
Rental share of housing stock in B	21.18%	22.56%	-1.38 pp
Universal transfer T	0.429	0.419	$+2.39\%$
Panel B. End-of-age-1 household balance sheets			
Liquid assets	6.54	0.09	$+6.45$
Housing wealth	18.97	11.37	$+7.60$
Mortgage debt	5.70	6.45	-0.75
Home equity	13.26	4.92	$+8.34$
Net financial assets	0.84	-6.36	$+7.20$

Notes: Effects are $B - B^0$; percentage changes use B^0 as the reference, and differences are calculated before rounding. Purchase prices are in RMB 10,000 per square meter; rents are the corresponding ten-year payments per square meter. Rental-stock shares measure floor space, not household population shares; their effects are percentage points. T is in RMB 10,000 per household per stage. Panel B reports within-age, end-of-stage means in RMB 10,000 per young household, including renters, whose housing wealth and home equity are zero. Net financial assets equal household liquid assets minus mortgage debt.

ownership-selection results in Section 6.2, this shows that the transfer economy accommodates lower-earning young homeowners who nevertheless use less leverage. Parental resources expand purchasing capacity beyond what the child's own earnings would support and allow a larger share of the purchase to be financed with equity. The conditional buyer statistics reflect both financing decisions and the composition of households selecting into Region A ownership.

Figure 7 traces the life-cycle pattern in average balance sheets. At ages 20–29, households in B hold more liquid assets and home equity with less mortgage debt. The home-equity and debt advantages persist through ages 30–49, when households receive no new inter vivos transfer. At ages 50–59, when households receive stochastic inheritance and choose transfers to their own children, liquid assets are lower in B than in B^0 and the home-equity gap narrows. At ages 60–79, the asset and home-equity profiles nearly coincide.

6.4 Distributional and Welfare Consequences

Transfers reduce housing inequality across young households' earning capacities while strengthening the association between housing outcomes and parental wealth. The Region A ownership gap between the top 20 percent and bottom 80 percent of child talent

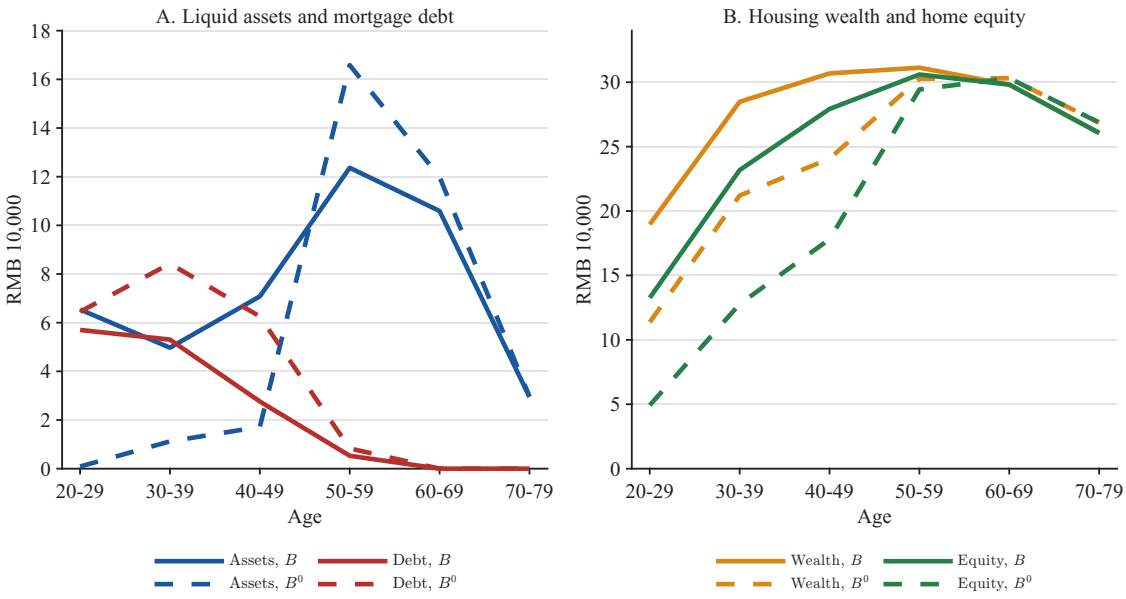


Figure 7: Life-Cycle Balance Sheets With and Without Family Transfers

Notes: Panel A reports liquid assets and mortgage debt; Panel B reports housing wealth and home equity. Solid lines denote B and dashed lines B^0. Series are within-age, end-of-stage population means in two stationary equilibria, measured in RMB 10,000. Zero mortgage debt in the final two age groups follows from the mortgage-contract assumptions.

narrows by 4.83 percentage points (Table 8). Parental support supplements children's own earnings in financing a purchase, reducing the ownership advantage associated with high earning capacity.

Access to this support, however, depends on parents' resources. Wealthier parents can provide more assistance, giving their children greater capacity to finance housing purchases. Using the benchmark parental-wealth cutoffs in both economies, the Region A ownership gap between children in the highest and lowest parental-wealth groups rises from 0.52 percentage points without transfers to 10.88 points with transfers.

I measure remaining-lifetime welfare using consumption-equivalent variation (CEV), the proportional adjustment to consumption and housing services in B^0 needed to match households' own welfare in B. Positive CEVs indicate gains from transfers. Appendix C.3 defines the welfare criterion and calculation.

Figure 8 shows that the welfare gains from transfers are concentrated early in life. The required compensation in B^0 is 26.33 percent for households aged 20–29, declines through middle age, and is approximately 3 percent for each age group from 50–59 onward. The all-living CEV is 11.60 percent.

The welfare response also varies substantially with talent. Among households aged 20–29, the CEV is 53.73 percent for the bottom talent quintile and declines to 8.66 percent for the 50th–80th percentiles. It becomes negative above the 80th percentile, reach-

Table 8: Distributional Consequences of Family Transfers

Outcome	B: transfers	B^0: no transfers	$B - B^0$
Panel A. Income, consumption, and net worth			
Labor-income Gini, ages 20–69	0.462	0.464	−0.001
Disposable-income Gini, all ages	0.469	0.470	−0.001
Consumption–housing composite Gini, all ages	0.342	0.386	−0.044
Net-worth p90/p50, all ages	3.67	4.68	−1.00
Share with nonpositive net worth, all ages	5.42%	12.84%	−7.42 pp
Panel B. Young-household housing inequality, ages 20–29			
Home-equity Gini	0.721	0.780	−0.058
Region A ownership gap: top-20 minus bottom-80 talent	11.03 pp	15.86 pp	−4.83 pp
Panel C. Parental-wealth gaps in young households' outcomes			
Region A ownership gap: parental Q4 minus Q1	10.88 pp	0.52 pp	+10.37 pp
Mean home-equity gap: parental Q4 minus Q1, RMB 10,000	22.98	0.08	+22.91

Notes: Effects are $B - B^0$, calculated before rounding. Ownership gaps are percentage points in every column. Labor-income inequality covers ages 20–69; the other Panel A measures pool all six age groups. Young-household home-equity inequality includes renters at zero. Net worth equals end-of-stage liquid assets plus home equity. Parental groups use beginning-of-age-4 net worth and the wealth cutoffs calculated in B, applied unchanged in B^0. Child outcomes are expected outcomes from the linked age-1 policies. Appendix C.2 gives the outcome definitions and grouping conventions.

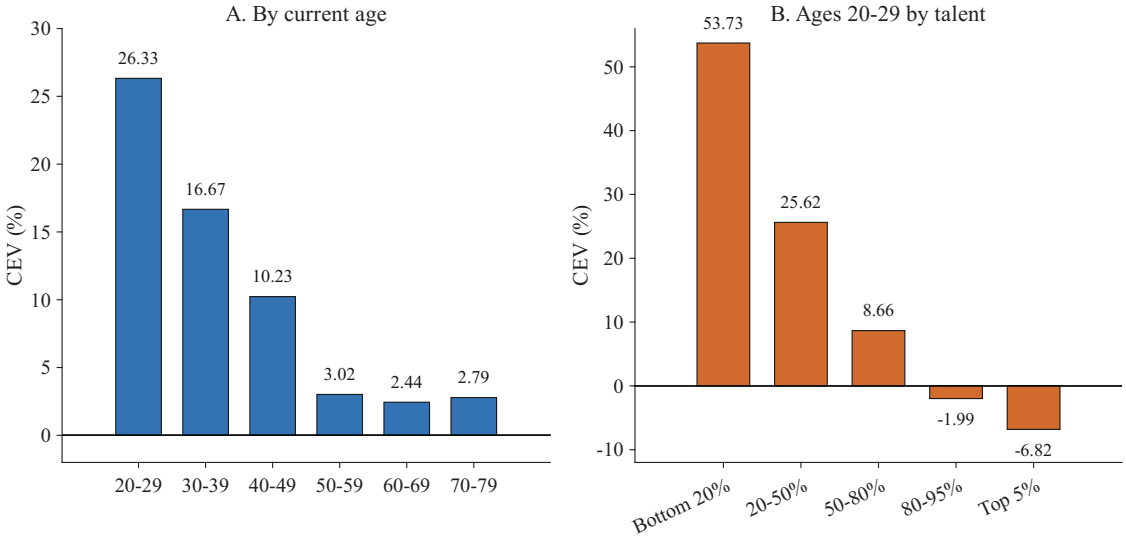


Figure 8: Remaining-Lifetime Welfare Effects of Family Transfers

Notes: Panel A reports consumption–housing CEVs by current age, and Panel B reports CEVs by talent among households aged 20–29. Positive values indicate the percentage increase in consumption and housing services required in B^0 to attain self-lifetime welfare in B; negative values indicate higher self-lifetime welfare in B^0. The comparison is between stationary equilibria. Appendix C.3 provides the welfare definition and calculation details.

ing -6.82 percent for the top 5 percent. High-talent households receive relatively little parental assistance in the benchmark. In the no-transfer economy, they face lower purchase prices and no longer devote resources to inter vivos transfers to their own children later in life.

Differences across talent remain pronounced at the parental stage. Although average welfare at ages 50–59 is higher with transfers, households in the top 5 percent of the talent distribution have a CEV of -19.67 percent (Online Appendix Figure S7), consistent with the larger amounts given by high-talent parents in B. The next section examines how public housing and place-based policies interact with these differences in family support and welfare incidence.

7 Public Policies and Private Family Finance

In this section, I evaluate two commonly used public policies for housing affordability, mortgage-credit expansion and rental assistance, and examine how they interact with the family transfer channel. LTV relaxation lowers the upfront financing requirement for purchase, while rental assistance subsidizes eligible households' rent payments in Region A.

Let B and B^0 denote the baseline equilibria with and without inter vivos family transfers. For a given policy, let P and P^0 denote the corresponding policy equilibria. I report the policy effects with and without private family finance,

$$\Delta_P^f(Y) = Y_P - Y_B, \qquad \Delta_P^0(Y) = Y_{P^0} - Y_{B^0},$$

and their difference,

$$I_P(Y) = \Delta_P^f(Y) - \Delta_P^0(Y).$$

The interaction $I_P(Y)$ indicates whether family transfers make the policy response in outcome Y larger or smaller. When useful, I also compare the effect of transfers in the baseline, $B - B^0$, with their effect in the policy economy, $P - P^0$.

Welfare comparisons use the self-lifetime consumption–housing CEV defined in Appendix C.3, with positive values indicating a loss relative to the corresponding baseline.

The first two columns of Table 9 summarize these principal experiments and their interactions with family finance.

Table 9: Policy Effects and Interactions with Family Transfers

Outcome	*Principal experiments*		*Additional experiments*	
	LTV relaxation	Rental assistance	Talent subsidy	Purchase voucher
Panel A. Policy effects with family transfers, $P - B$				
Spending / tax revenue (%)	0.00	2.14	18.06	10.85
$\Delta \Pr(A)$, age 1 (pp)	0.096	1.636	0.911	0.408
$\Delta \Pr(oA)$, age 1 (pp)	0.128	-0.350	0.813	0.460
$\Delta[\Pr(oA) + \Pr(oB)]$, age 1 (pp)	0.053	-1.505	0.108	0.138
Δ aggregate output (%)	0.016	0.034	0.178	0.144
All-living welfare cost, CEV (%)	0.74	0.43	2.22	1.46
Panel B. Policy–transfer interactions				
$I_P[\Pr(A)]$, age 1 (pp)	-0.021	-0.976	0.157	-0.151
$I_P[\Pr(oA)]$, age 1 (pp)	-0.338	-0.072	0.006	-0.040
I_P[home equity], age 1	0.399	-0.167	0.587	0.288

Notes: Panel A reports $P - B$. Panel B reports $I_P = (P - B) - (P^0 - B^0)$. Probability effects are percentage points. Home equity is an age-1 mean in units of RMB 10,000. A positive CEV means that the policy economy requires compensation to attain welfare in its corresponding baseline. CEVs are pairwise and are not differenced to form an interaction. The LTV experiment has no direct fiscal expenditure in the model, although financial-stability costs are outside the model.

7.1 LTV Relaxation: Formal Credit versus Family Finance

The LTV reform tests whether formal mortgage credit can substitute for parental support at the purchase-entry margin and whether family resources continue to shape leverage and home equity after purchase.

Minimum down-payment regulation is a central instrument of Chinese housing finance. In 2024, national regulators lowered the minimum down-payment floor for commercial mortgages, while allowing local authorities some discretion over implementation (People's Bank of China and National Financial Regulatory Administration, 2024). I study a stronger, stylized reform that raises the maximum LTV ratio from 70 to 90 percent for new purchases in both regions. The reform lowers the minimum down payment from 30 to 10 percent. Existing mortgages are not refinanced, and the payment-to-income limit, mortgage rate, maturity, and amortization schedule remain unchanged.

Relaxing the LTV ceiling mainly changes how young households finance ownership. Among age-1 Region A buyers, mean origination LTV rises from 44.1 to 54.4 percent, while the age-1 Region A ownership share increases by only 0.13 percentage points. Without parental transfers, the ownership response is larger, at 0.47 points. Formal credit therefore substitutes for family resources at purchase entry, while parental resources continue to shape debt and equity after entry.

Figure 9 shows that family transfers continue to reduce young households' mortgage debt and raise home equity under the higher ceiling. These balance-sheet means cover all

young households, including renters, and reflect both financing decisions and selection into ownership; the LTV comparison above is conditional on Region A purchase.

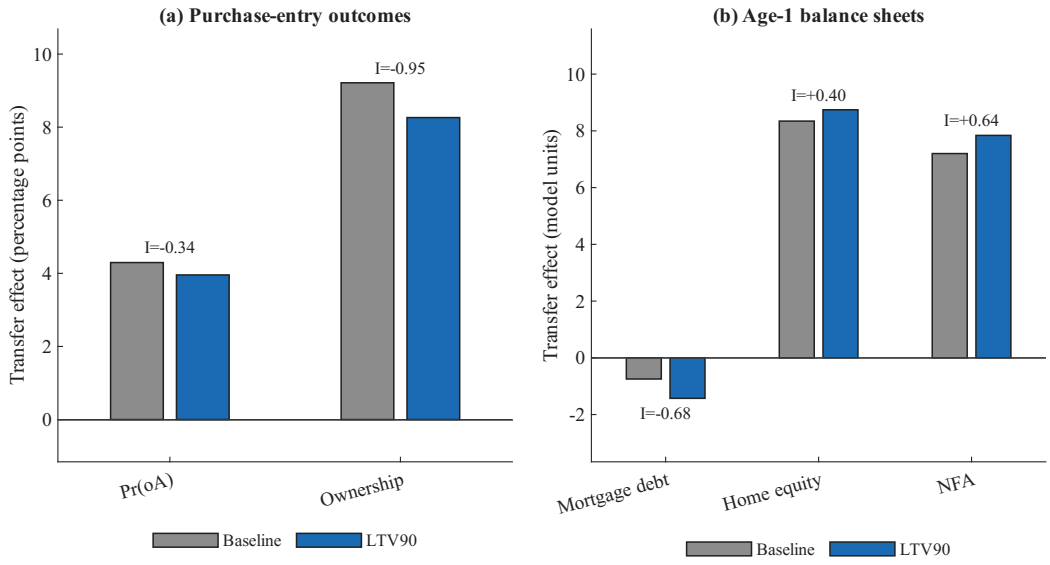


Figure 9: Family Transfers under Relaxed LTV Regulation

Notes: Gray bars report the effect of family transfers in the baseline, $B - B^0$, and blue bars report their effect under LTV90, $P - P^0$. Labels give the policy–transfer interaction $I_P = (P - P^0) - (B - B^0)$. Panel (a) reports percentage points; Panel (b) reports means over all age-1 households, including renters, in units of RMB 10,000.

Under the self-lifetime consumption–housing criterion, the reform produces an all-living welfare cost of 0.74 percent relative to the baseline, incorporating the equilibrium adjustment of housing markets and household allocations (Online Appendix Section S4.1).[2]

7.2 Rental Assistance: Spatial Access without Ownership

Rental assistance instead targets spatial access. I use it to assess whether public support reduces young households' dependence on family wealth for residence in the productive region while parental resources remain important for homeownership there.

China has promoted subsidized rental housing and monetary rental assistance as ways to broaden access to urban housing (General Office of the State Council of China, 2016). I consider a program for households at ages 1 and 2 whose potential annual Region A income is no higher than the baseline 60th percentile of annual income among Region A residents at ages 1 and 2 and who own no house at the beginning of the period. Eligible households renting in Region A receive a 20 percent discount on market rent for up to 70 square meters. Landlords receive the full market rent, the government pays the

[2]The model abstracts from mortgage default, lender losses, risk-based pricing, and financial-stability consequences.

difference, and eligibility is reassessed in each period. Among eligible households, 23.6 percent take up the subsidy at age 1 and 21.7 percent at age 2. Total expenditure equals 2.14 percent of tax revenue.

Rental assistance subsidizes eligible households' rent payments in Region A without subsidizing a home purchase. The age-1 Region A residence share rises by 1.64 percentage points with family transfers and by 2.61 points without them (Figure 10). The Region A rental share increases, while ownership shares fall in both regions. These net changes in age-1 location–tenure shares show expanded access through renting alongside lower homeownership.

The larger residence response without family transfers indicates that rental assistance substitutes for parental resources in access to Region A. Its effect on the ownership role of family finance is much smaller: the difference in age-1 Region A ownership between the transfer and no-transfer economies remains nearly unchanged. Public rental support thus provides an alternative route into the productive labor market while leaving the role of family resources in ownership largely intact.

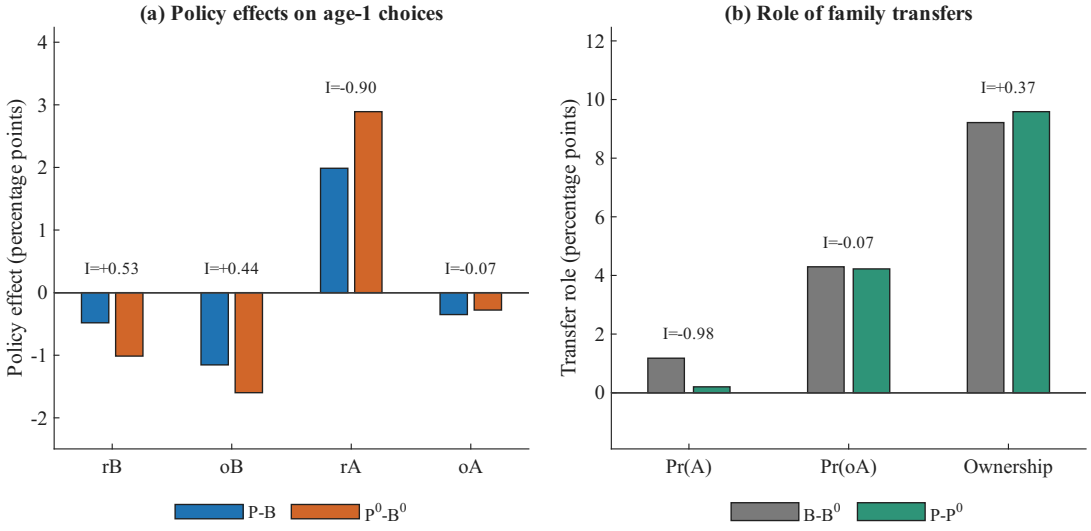


Figure 10: Affordable Renting and the Role of Family Transfers

Notes: Panel (a) reports net changes in age-1 location–tenure shares with and without family transfers. The higher Region A rental share is accompanied by lower shares in both Region B choices and Region A ownership. These are equilibrium share differences, not measured household transitions. Panel (b) compares the effect of family transfers in the baseline and rental-policy economies. Rental assistance substantially reduces the role of transfers in Region A residence while leaving their role in Region A ownership largely intact. Labels report the policy–transfer interaction. Effects are percentage points.

Higher rental demand raises the Region A market rent by 2.99 percent, partly offsetting the statutory discount. The larger working-age population in Region A raises effective labor, although lower mean age-adjusted productivity partly offsets this gain. Aggregate output rises only slightly. Actual rental-assistance recipients aged 20–29 gain 1.65

percent in remaining-lifetime consumption-equivalent welfare. Income-ineligible age-1 households receive no direct subsidy and lose on average. The all-living welfare cost is 0.43 percent (Online Appendix Section S4.2).

I also examine two policies for households aged 20–39 in the top talent quintile, a RMB 300,000 subsidy for eligible Region A residence and a voucher covering 20 percent of an eligible A home purchase, each available once per lifetime. Their effects on young households' Region A ownership differ little between economies with and without family transfers (Table 9), consistent with the concentration of transfer-induced ownership gains among lower-earning households. Both policies increase Region A's effective labor even as its all-age population share falls; the voucher also encourages larger housing purchases, with little increase in overall homeownership (Online Appendix Sections S4.3 and S4.4).

8 Conclusion

CHIP evidence highlights a distinction between residence in productive cities and ownership within them. The quantitative analysis shows that family transfers primarily reshape housing tenure and the distribution of housing wealth, with more modest effects on productive-city residence and aggregate output. Rental housing allows young workers to access productive labor markets without meeting the financing requirements for purchase. At the same time, modest transfers can make ownership in lower-cost cities more attractive relative to renting in the productive region. Family assistance therefore changes both tenure within a location and the relative attractiveness of purchasing across locations, rather than uniformly encouraging residence in productive cities. In equilibrium, transfers increase the productive region's working population, but the accompanying decline in average age-adjusted productivity partly offsets the contribution of this population expansion to effective labor.

Parental support expands ownership principally among young households with lower own earning capacity. In the transfer economy, young homeowners in the productive region have lower mean labor earnings but also use less mortgage leverage. Transfers narrow the ownership gap across children's talent groups and reduce inequality in young households' home equity. At the same time, wealthier parents provide larger amounts of assistance, widening the differences in children's ownership and equity across parental-wealth groups. The expansion of ownership among lower-earning young adults therefore coexists with a stronger relationship between housing outcomes and family background.

Family transfers raise aggregate welfare, with gains concentrated among young and lower-talent households. Assistance at the beginning of adult life supports consumption and liquidity and affects subsequent saving, mortgage repayment, and housing-equity accumulation. Some high-talent households nevertheless have lower self-welfare in the transfer economy: they receive comparatively little assistance when young, face higher purchase prices, and devote resources to their own children later in life.

Households that receive no parental transfers are also affected by the resulting changes in house prices and rents. Transfer-financed purchase demand raises equilibrium house prices, increasing the value of existing housing holdings and the acquisition costs faced by prospective buyers without parental assistance. In the benchmark, households that access Region A's labor market through renting face slightly lower rents per square meter.

The policy experiments distinguish the roles of public support and family resources. Relaxing mortgage borrowing limits partly substitutes for parental funds in financing home purchase, while family transfers continue to affect young households' leverage and equity. Rental assistance reduces dependence on parental support for residence in the productive region without materially reducing its role in ownership.

Note on the appendices. Appendix A documents the model structure and solution method; Appendix B describes the data and calibration; Appendix C defines outcome measures, decompositions, and welfare; and Appendix D reports sensitivity and numerical accuracy checks. The online appendix provides expanded baseline, no-transfer, and policy results, additional policy experiments, and numerical implementation details. The main text presents the economic arguments and principal findings and can be read independently of these supporting materials.

Appendix A Model Details and Numerical Solution

This appendix records the household transitions, inheritance construction, housing-supply system, and aggregation conventions.

A.1 Household Problems and Feasibility Constraints

The renter, buyer, and continuing-owner budgets are given in equations (17)–(20). Their housing and mortgage transitions are

$$\begin{aligned} j = r\ell: &\quad (\ell', \pi') = (\ell, r), && h' = m' = q' = 0, \\ j = o\ell: &\quad (\ell', \pi', h') = (\ell, o, h), && (m', q') = (\vartheta_{K_g} M, \max\{K_g - 1, 0\}), \\ j = k\ell_-: &\quad (\ell', \pi', h') = (\ell_-, o, h_-), && (m', q') = (\vartheta_q m, \max\{q - 1, 0\}). \end{aligned}$$

Renting resets the housing and mortgage states. Selling realizes depreciated home value net of outstanding debt; keeping permits no moving, resizing, selling, or refinancing. All choices require $c > 0$ and $x' \geq 0$, and renters and new buyers satisfy $h \geq \underline{h}$. Renters cannot borrow. New mortgages satisfy both the LTV and PTI restrictions in equation (19); family transfers are excluded from the gross-income base of the PTI limit. Terminal-age households cannot originate mortgages, and their warm-glow estates must be nonnegative.

An entrant starts without liquid assets, housing, or debt and receives the parental transfer f. At ages 40–49, every feasible action integrates over inheritance ι and child talent z_c, observed upon entry into ages 50–59. The resulting state is

$$s^4(j) = \big(z, x' + \iota/R_s, \ell'(j), \pi'(j), h'(j), m'(j), q'(j); z_c\big).$$

Inheritance enters beginning liquid assets once. At the parental stage, $f \geq 0$ reduces current resources and enters the altruistic term $\alpha_p \overline{V}^1(z_c, f, \ell_-)$. The child's origin is the parent's beginning-of-period region, irrespective of the parent's current location choice. These rules apply to renters, new buyers, and continuing owners (Online Appendix Section S1.1).

A.2 Mortgage Amortization Schedule

The mortgage contract is fully amortizing with level monthly payments. Let r_m^{ann} denote the annual mortgage rate and $i_m = r_m^{\text{ann}}/12$ the corresponding monthly rate. At the beginning of a model period, $q \in \{0, 1, 2, 3\}$ denotes the number of ten-year payment blocks remaining. The mortgage-free state has $q = 0$, zero outstanding debt $m = 0$, and zero scheduled payments, $a_0(0) = \Pi_0(0) = 0$. Without new borrowing, its next-period mortgage state remains $m' = q' = 0$. The amortization formulas below apply only to $q \geq 1$, for which $120q$ monthly payments remain. For a beginning balance m, the level monthly

payment is

$$a_q(m) = m\frac{i_m}{1-(1+i_m)^{-120q}}. \tag{A.1}$$

Because one model period comprises 120 months, the mortgage payment entering the household's ten-year budget is

$$\Pi_q(m) = 120a_q(m) = \pi_q m, \qquad \pi_q = \frac{120i_m}{1-(1+i_m)^{-120q}}. \tag{A.2}$$

Thus, π_q is the total scheduled principal and interest paid during the current ten-year period per unit of beginning mortgage balance; it is not an interest rate.

After these 120 monthly payments, the remaining balance is

$$m' = \vartheta_q m, \qquad \vartheta_q = \frac{(1+i_m)^{120q}-(1+i_m)^{120}}{(1+i_m)^{120q}-1}. \tag{A.3}$$

The mortgage state then moves from q to $q-1$. In particular, $\vartheta_1 = 0$ because a mortgage with one block remaining is fully repaid within the current model period.

With $r_m^{\text{ann}} = 0.0501$, and hence $i_m = 0.004175$, Table A.1 reports the coefficients used in the baseline.

Table A.1: Mortgage payment and remaining-balance coefficients

Remaining blocks q	Remaining maturity	π_q	ϑ_q
1	10 years	1.273373	0.000000
2	20 years	0.792610	0.622449
3	30 years	0.644920	0.813666

Notes: π_q aggregates the next 120 monthly payments relative to the beginning balance. ϑ_q is the fraction of that balance outstanding after the same payments.

A.3 Stochastic Inheritance Distribution

The common inheritance distribution is constructed from the mass-weighted distribution F_e of age-6 terminal estates in equation (24). Its five nodes are conditional means within weighted-quantile bins,

$$b_k = \frac{1}{p_k^b}\int_{q_{k-1}}^{q_k} F_e^{-1}(u)\,du, \qquad p_k^b = q_k - q_{k-1}, \qquad F_e^{-1}(u) = \inf\{e : F_e(e) \geq u\}.$$

The bin edges are $(0, 0.25, 0.50, 0.75, 0.90, 1)$. Their widths give fixed inheritance probabilities $(0.25, 0.25, 0.25, 0.15, 0.10)$. Discrete estate mass crossing a bin boundary is split

between adjacent bins at the same estate value, preserving these probabilities. The baseline nodes are

$$(b_1, \ldots, b_5) = (7.5371, 12.5762, 20.3507, 36.7314, 97.3737)$$

in units of 10,000 RMB.

A household receives $\iota = b_k$ with probability p_k^b once, upon entry into age group $g = 4$. Since beginning liquid wealth earns R_s, the transition is $x_4 = x_3' + b_k/R_s$, increasing age-4 resources by exactly b_k. Inheritance is a one-time resource shock, not a persistent state. The draw is independent of the recipient's productivity and earlier family-transfer history and is not linked to a particular parent's estate. It preserves the stationary marginal estate distribution without a separate dynastic-wealth state.

At the stationary fixed point, each candidate node b_k equals the conditional mean estate in the same quantile bin generated by the household choices and stationary distribution under those nodes. The probabilities remain fixed (Online Appendix Section S1.2).

A.4 Housing Supply and Owner–Rental Allocation

Owner and renter floor-space demands are

$$D_{\ell,t}^{\text{own}} = \sum_{g=1}^{6} \int h \, d\mu_{o\ell,t}^{g}, \qquad D_{\ell,t}^{\text{rent}} = \sum_{g=1}^{6} \int h \, d\mu_{r\ell,t}^{g}. \tag{A.4}$$

At the reference equilibrium, I back out total floor space and its tenure shares as

$$H_{\ell,0} = D_{\ell,0}^{\text{own}} + D_{\ell,0}^{\text{rent}}, \qquad s_{\ell,0}^{r} = \frac{D_{\ell,0}^{\text{rent}}}{H_{\ell,0}}, \qquad \Omega_{\ell,0} = 1 - s_{\ell,0}^{r}. \tag{A.5}$$

Here $\Omega_{\ell,0}$ is the owner share of floor space, not the household homeownership rate. Stationarity gives the construction shifter

$$\bar{I}_\ell = \frac{\delta_h H_{\ell,0}}{P_{\ell,0}^{\rho_\ell}}. \tag{A.6}$$

These backouts reproduce all four reference segment demands at $(P_{\ell,0}, R_{\ell,0})$.

The rental-share rule in equation (28) implies

$$Q_{\ell,t} \equiv \left[\frac{R_{\ell,t}/P_{\ell,t}}{R_{\ell,0}/P_{\ell,0}}\right]^{\chi_\ell}, \qquad s_{\ell,t}^{r} = \frac{s_{\ell,0}^{r} Q_{\ell,t}}{1 - s_{\ell,0}^{r} + s_{\ell,0}^{r} Q_{\ell,t}}. \tag{A.7}$$

Thus, χ_ℓ is the elasticity of the rental-to-owner stock ratio with respect to the rent-to-price ratio. With $H_{\ell,t} = \bar{I}_\ell P_{\ell,t}^{\rho_\ell}/\delta_h$, segment supplies are

$$S_{\ell,t}^{\text{rent}} = s_{\ell,t}^{r} H_{\ell,t}, \qquad S_{\ell,t}^{\text{own}} = (1 - s_{\ell,t}^{r}) H_{\ell,t}. \tag{A.8}$$

The purchase price changes both total stock and its tenure allocation; rent changes the allocation without directly changing total stock. Market clearing requires $D_\ell^{\text{own}} = S_\ell^{\text{own}}$ and $D_\ell^{\text{rent}} = S_\ell^{\text{rent}}$ in each region. The backed-out intercepts and reference shares remain fixed in policy and no-transfer equilibria (Online Appendix Section S1.5).

A.5 Numerical Solution and Convergence

Given prices, wages, the universal transfer, and inheritance nodes, I solve the six household problems on finite grids. The parent's altruistic term links age-4 choices to the age-1 value function, requiring a fixed point in household values. The induced policies determine the stationary distribution and terminal estates. In general equilibrium, I update (P_A, P_B, R_A, R_B, T) and inheritance nodes until the four housing markets, government budget, and inheritance distribution are consistent.

Table A.2 summarizes the baseline grids. Assets, housing, and transfers use nonuniform nodes concentrated near financing margins, and the productivity grid has a probability-weighted mean of one. The state space also includes location–tenure status, parental origin for entrants, and age-specific mortgage states.

Table A.2: Baseline Numerical Grids

Grid object	Number of nodes	Range or discrete values
Productivity, z	13	$[0.3003,\ 12.3146]$
Liquid assets, x and x'	10	$[0,\ 500]$
Housing size, h	13	$[10,\ 200]$
Inter vivos transfer, f	13	$[0,\ 300]$
New mortgage as a fraction of the LTV ceiling	5	$\{0, 0.25, 0.50, 0.75, 1\}$
Mortgage duration, q	4	$\{0, 1, 2, 3\}$ ten-year blocks
Inheritance, b	5	endogenous; $[7.5371,\ 97.3737]$ in the baseline

Notes: Liquid assets, transfers, and inheritance are in units of 10,000 RMB; housing is in square meters. The new-mortgage grid is defined over $M/(\theta_{\text{LTV}} P_\ell h)$: origination debt equals the listed fraction multiplied by the LTV ceiling. Candidates that violate the PTI limit are excluded; the grid is not rescaled to that limit.

The value-function and stationary-distribution tolerances are 10^{-5} and 10^{-4}, respectively. The normalized inheritance discrepancy and maximum absolute relative housing and fiscal gaps have nominal tolerances of 10^{-3}; the government budget is also checked

in levels. For policy and sensitivity cases with persistent housing-residual oscillation, I allow housing gaps up to 1.5×10^{-3} under the stopping rule documented in Online Appendix Section S1.5, while keeping the other criteria unchanged. The fixed-tenure-share sensitivity is separately accepted at 1.474×10^{-3} without satisfying that oscillation rule. The tighter GE check retains its 5×10^{-4} threshold without an exception. Actual residuals and effective tolerances are reported for each accepted equilibrium.

A.6 Aggregation Conventions

Saved within-age distributions have mass one. To construct full-population measures, each receives the age weight $1/6$ once; each resulting μ^g already has mass $1/6$ across both regions. Within-age statistics normalize by the relevant age mass, whereas population aggregates use these weighted measures. Only ages 1–5 supply labor, giving

$$L_\ell = \sum_{g=1}^{5} \int G_g z \, d\mu_\ell^g(z), \tag{A.9}$$

$$Y_\ell = \mathrm{TFP}_\ell L_\ell, \qquad Y = Y_A + Y_B. \tag{A.10}$$

Output here is productivity-normalized. Monetary ratios instead use $Y^{\mathrm{mon}} = wY = 10w^{\mathrm{ann}}Y = 50Y$, in units of 10,000 RMB over a ten-year stage. Spending, family transfers, and wealth are divided by this monetary output; wealth/output ratios therefore use ten-year, not annual, output. The scale cancels in percentage output changes.

The fixed age and talent distributions and inelastic labor supply fix total effective labor; location choices determine its regional allocation. The financial sector elastically absorbs liquid assets and supplies mortgage credit at externally fixed terms, so there is no bond-market-clearing condition. Liquid assets net of mortgage debt are reported as net financial assets. Government expenditure comprises pensions, the universal transfer, and policy outlays, financed by taxes on working-age gross labor income. Family transfers and inheritances are private flows.

Appendix B Data and Calibration

The quantitative analysis uses the 2017 China Household Finance Survey (CHFS). I use individual records to construct the regional population shares and earnings process, and household records to construct tenure, purchase-price, and rent moments. All CHFS statistics use the corresponding individual or household survey weights. CHIP 2018 supplies the unweighted parental housing assistance receipt target described in Ap-

pendix B.6. This appendix describes the samples and variable construction, then maps the resulting moments into the calibrated objects.

B.1 Survey, Geography, and Samples

The CHFS reports demographic characteristics, labor income, education, location, housing tenure, dwelling size, self-assessed home value, and rent. I merge the individual and household files with the survey's city identifiers and weights. The analysis excludes rural observations because rural housing is often obtained through village-allocated homestead land rather than the market prices and mortgage contracts represented in the model. The restriction also aligns the model's talent-based earnings process with urban labor markets; the model does not represent land-based rural production.

Region A comprises 18 Tier 1 and Tier 2 cities: Beijing, Shanghai, Guangzhou, Shenzhen, Chengdu, Hangzhou, Wuhan, Chongqing, Nanjing, Tianjin, Suzhou, Xi'an, Changsha, Shenyang, Qingdao, Zhengzhou, Dalian, and Ningbo. Region B contains the remaining urban cities. Age is survey year minus birth year. The first five empirical bins are 20–29 through 60–69; the terminal bin pools respondents aged 70 and older and maps to the model's sixth life-cycle stage.

Regional population shares and the earnings process use individual records and individual weights. Tenure, purchase prices, and rents use the household head, the household housing record, and household weights. The estimation sample for the working-age earnings process contains 24,465 individuals with positive labor income and valid age, region, education, and survey weight. The final purchase-price samples contain 8,401 owner households in Region A and 10,626 in Region B; the rent samples contain 2,165 and 1,470 renter households, respectively. Sample sizes differ because each moment requires its own valid income or housing fields.

B.2 Location and Tenure Targets

I first compute the weighted share of individuals living in Region A within each age group. The raw shares are 31.8, 33.2, 29.4, 30.9, 34.9, and 34.5 percent from the youngest to the oldest group. I combine these estimates with weighted homeownership rates among household heads in each age–region cell. Because some conditional tenure cells are small, the calibration uses rounded life-cycle profiles rather than treating sampling variation in every cell as a separate target. Table B.1 reports the three profiles used to construct the 24 joint location–tenure targets.

Let A_g denote the target Region A share and let $O_{\ell g}$ denote the target ownership rate

conditional on region ℓ at age g. The four target shares are

$$s_{rB,g} = (1 - A_g)(1 - O_{Bg}), \qquad s_{oB,g} = (1 - A_g)O_{Bg},$$
$$s_{rA,g} = A_g(1 - O_{Ag}), \qquad s_{oA,g} = A_g O_{Ag}.$$

This construction makes the regional and conditional-tenure content of the targets explicit and guarantees that the four shares sum to one within age.

Table B.1: Construction of Age-Specific Location–Tenure Targets

	Target components			Implied joint shares			
Age	$\Pr(A)$	$\Pr(o \mid B)$	$\Pr(o \mid A)$	rB	oB	rA	oA
20–29	35.0	60.0	37.0	26.00	39.00	22.05	12.95
30–39	35.0	85.0	60.0	9.75	55.25	14.00	21.00
40–49	30.0	85.0	70.0	10.50	59.50	9.00	21.00
50–59	30.0	85.0	70.0	10.50	59.50	9.00	21.00
60–69	35.0	85.0	70.0	9.75	55.25	10.50	24.50
70 and older	35.0	85.0	70.0	9.75	55.25	10.50	24.50

Notes: Entries are percentages. The component profiles are rounded benchmarks constructed from CHFS-weighted region and tenure profiles. The implied shares are the targets reported in Table 3. The terminal survey group maps to the model stage labeled 70–79.

B.3 Earnings, Regional Productivity, and Talent

Individual labor income is the sum of the two reported monthly labor-income components. I drop observations for which both components are missing or refused, set a single missing component to zero, and retain positive total income. Monthly income is converted to annual units of 10,000 RMB and winsorized at the first and ninety-ninth percentiles before estimation. No additional trimming is applied in the earnings regressions, which use the individual survey weights and working ages 20–69.

With mean productivity normalized to one, I set $w^{\text{ann}} = 5$, or RMB 50,000 before age and regional adjustments; the ten-year model wage is $w = 50$.

To recover the age profile and permanent talent distribution, I estimate

$$\log y_i = a + \sum_{g \neq 2} d_g \mathbb{1}\{g_i = g\} + d_A \mathbb{1}\{\ell_i = A\} + b_e \,\mathrm{edu}_i + u_i, \tag{B.1}$$

where education categories are mapped into years of schooling and ages 30–39 are the reference group. Exponentiating the age coefficients gives the age-efficiency schedule in Table B.5; I set $G_6 = 0$ because the last stage is retired. Permanent talent retains both the

schooling component and the residual. Specifically, I remove only the constant, age effect, and region effect from log income, exponentiate, and normalize the weighted mean of z_i to one.

A weighted maximum-likelihood fit of $F(z) = \exp[-(s_z/z)^{\theta_z}]$ to these normalized observations gives $\theta_z = 1.9266$ and $s_z = 0.5430$. The baseline uses 13 conditional-bin means: nine decile bins through the ninetieth percentile and four upper-tail bins with probabilities 0.05, 0.03, 0.01, and 0.01. The grid spans 0.3003 to 12.3146 and is renormalized so that its probability-weighted mean is exactly one. Productivity remains fixed over the life cycle, and the benchmark assigns each heir an independent draw from this distribution. Appendix D.3 allows positive intergenerational persistence in talent.

A separate weighted log-earnings regression with age-group and Region A indicators provides the regional earnings premium used to set $\mathrm{TFP}_A = 1.37$ relative to $\mathrm{TFP}_B = 1$.

The taste-shock scale is then calibrated to an equilibrium working-age gross-income ratio of 2.0 between Regions A and B. This ratio is a sorting target, not the raw regional mean in the cleaned earnings sample: the latter is 1.404, while the age-and-education-adjusted regional coefficient in (B.1) implies 1.275. Keeping these objects separate prevents the exogenous regional productivity premium from absorbing the endogenous sorting governed by σ_ν.

I set the pension to one-half of the location-independent earnings base wG_4z, corresponding to $\rho_{\mathrm{pen}} = 0.50$. This replacement rate is close to the 46 percent average pension replacement ratio reported for China in 2017 by Fang and Feng (2020) and lies within the 40–60 percent adequacy range in IMF (2022).

The labor-income tax is proportional: all working-age households pay 10 percent of gross labor earnings. This assigned rate, $\tau_l = 0.10$, remains fixed across counterfactuals.

Appendix Figure B.1 shows the weighted distributional fit used to construct the productivity grid.

B.4 Productivity Sorting across Location and Tenure

I use the productivity gap between young Region A renters and Region B owners as an approximate calibration target for the recurring rental friction. The main sample consists of urban household heads aged 20–29 with positive labor income and valid productivity, tenure, and household weights. The alternative sample assigns the household's tenure to every positive-income worker aged 20–29 and uses individual survey weights. Table B.2 reports both definitions together with the stationary age-1 model means.

The stable empirical restriction is that Region A renters are more productive than Region B owners. The difference is 9.0 percent among household heads and 28.2 percent in

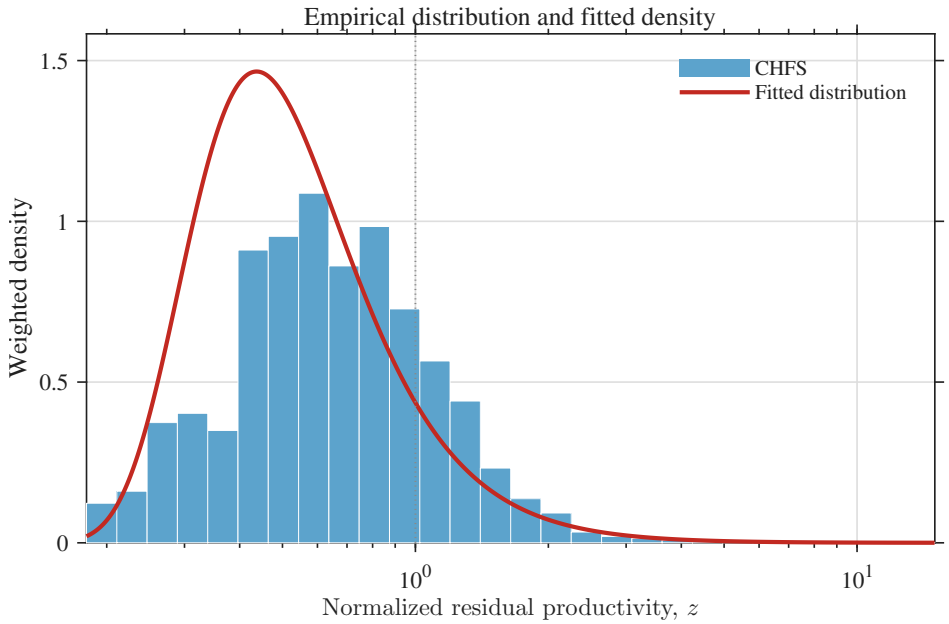


Figure B.1: Residual Productivity Distribution and Fréchet Fit

Notes: The figure plots the CHFS-weighted distribution of normalized residual productivity and the fitted mean-one Fréchet density. The dotted vertical line marks $z = 1$.

Table B.2: Young-Household Productivity by Location and Tenure

Choice	CHFS household heads	CHFS all workers	Baseline model
rB	1.407 (114)	1.112 (355)	0.924
oB	1.367 (211)	1.008 (2,271)	0.908
rA	1.490 (322)	1.291 (662)	1.012
oA	1.529 (192)	1.078 (1,336)	1.391

Notes: Entries are survey-weighted or stationary-mass- weighted conditional mean talent. Unweighted CHFS sample sizes are in parentheses. CHFS talent is normalized to have mean one in the full working-age estimation sample; the model talent grid has an ex ante mean of one. Conditional productivity levels are untargeted. The rA–oB percentage gap in the household-head sample is targeted approximately; the all-worker sample provides a robustness check.

the all-worker sample, while the model implies 11.4 percent. I target the direction and approximate size of the household-head gap; the all-worker estimate provides a robustness check. The sensitivity to the unit of observation motivates an approximate fit rather than an exact match. Appendix D.5 shows how this sorting moment changes when the recurring rental friction is removed.

Mobility-cost schedules. The CHFS does not separately identify the resource cost of entering Region A from unobserved location preferences. I assign entry costs externally, while the age-specific location–tenure constants absorb residual differences in nonpecuniary attractiveness. Morten and Oliveira (2016) estimate that the mean observed cost of

moving between municipalities is 0.8–1.2 times the mean annual wage, with most of the cost taking the form of a fixed component. The entry schedule in Table B.5 equals 0.8–1.8 times baseline annual earnings, $w^{\text{ann}} = 5$. The first three age groups span the estimated range; the increasing profile at later ages follows evidence that moving costs rise with age (Kennan and Walker, 2011). The cost is paid only when a household whose previous region—or inherited origin at entry—is Region B chooses either rA or oA.

The recurring Region A rental-friction schedule in Table B.5 equals 0.4–0.6 times baseline annual earnings. It is lower than the entry cost at every age and is paid only when a household chooses rA. It summarizes recurring resource costs of renting in Region A, motivated by unequal access to housing-linked local public services (Zhang and Chen, 2018, General Office of the State Council of China, 2016, General Office of the Guangzhou Municipal Government, 2017). I set this schedule separately from the jointly fitted parameters to match the direction and approximate size of the household-head productivity gap described above.

B.5 Purchase Prices and Rents

For owner households, I use the reported value and floor area of the primary residence. I replace interval-coded values by their bracket midpoints and winsorize reported values at the weighted first and ninety-ninth percentiles. I remove floor-area observations above the weighted 99.5th percentile because a small number of reported dwellings have extremely large areas and would otherwise distort the regional price moments. The regional purchase-price moment is the household-weighted mean of each owner's unit price,

$$P_{\ell}^{\text{data}} = \frac{1}{10{,}000} \frac{\sum_{i \in \mathcal{O}_{\ell}} w_i (V_i / H_i)}{\sum_{i \in \mathcal{O}_{\ell}} w_i}. \tag{B.2}$$

Here $\ell \in \{A, B\}$ indexes the region, $\mathcal{O}_{\ell}$ is the set of sampled owner households in region ℓ, w_i is household i's CHFS sampling weight, V_i is the winsorized value of its primary residence in RMB, and H_i is its floor area in square meters. The factor $1/10{,}000$ converts the resulting unit price into 10,000 RMB per square meter. This yields $P_A = 1.8373$ and $P_B = 0.4747$ in units of 10,000 RMB per square meter, or RMB 18,373 and RMB 4,747 per square meter.

For renter households, I require positive monthly rent and floor area and apply the same weighted 99.5th-percentile floor-area trim. The model-period rent per square meter

is

$$R_\ell^{\text{data}} = \frac{10 \times 12}{10{,}000} \frac{\sum_{i \in \mathcal{R}_\ell} w_i (\text{rent}_i^{\text{mo}} / H_i)}{\sum_{i \in \mathcal{R}_\ell} w_i}. \tag{B.3}$$

The resulting ten-year rents are $R_A = 0.3108$ and $R_B = 0.1576$ in model units. For reference, the corresponding weighted mean dwelling sizes are 99.34 and 123.16 square meters for owners in Regions A and B, and 53.80 and 65.85 square meters for renters. These size statistics are descriptive and are not calibration targets.

I map these four price and rent moments into the reference housing-supply system using model-implied regional floor-space demand. The owner-space share $\Omega_{\ell,0}$ is owner-occupied floor space divided by total regional floor space, not the household homeownership rate. The backout gives $\Omega_{A,0} = 0.551$ and $\Omega_{B,0} = 0.788$, corresponding to reference rental shares of 0.449 and 0.212. Given $\delta_h = 0.140$ and the assigned supply elasticities, the stationary stock condition gives $\bar{I}_A = 1.381$ and $\bar{I}_B = 9.686$. These objects reproduce owner and renter floor-space demand at the four target prices and rents; Appendix A.4 provides the backout equations and segment-market residuals.

The supply elasticities are assigned rather than estimated from this sample.[3]

Annual housing depreciation is 1.5 percent, consistent with the benchmark in Kaplan et al. (2020), giving a ten-year rate of $\delta_h = 1 - (1 - 0.015)^{10} = 0.140$. The minimum dwelling size, $\underline{h} = 10$ m^2, is about one quarter of average urban per-capita housing space in the CHFS.

B.6 Parental Housing Assistance and Altruism

The 2018 China Household Income Project (CHIP) urban household and person files provide the parental-assistance moment used to calibrate α_p. I merge the files by household identifier and retain households whose heads are aged 20–29, with valid current-tenure and parental-housing-assistance responses. The sample contains 615 households. I use the same Region A–B classification as in Section 2; the target pools homeowners in both regions. All CHIP moments are unweighted.

The housing question records parental contributions to buying or building housing in brackets. I classify the three positive-contribution brackets as assistance and the no-contribution response as no assistance. Of the 438 current homeowners, 275 report assis-

[3]Saiz (2010) estimates metropolitan housing-supply elasticities using demand variation, terrain and water coverage, and land-use regulation. Mabille (2023) uses these estimates in a quantitative spatial housing model.

tance. The calibration target is therefore

$$\Pr(f > 0 \mid g = 1,\ \ell \in \{A, B\},\ \text{owner}) = \frac{275}{438} = 0.6278539.$$

The denominator includes current homeowners only. The 307 recipients among all 615 young households instead imply an unconditional incidence of 49.9 percent, which is not the calibration target. Strict renters, free accommodation, and other tenure categories remain distinct in the descriptive analysis.

I jointly fit α_p, the 18 location–tenure constants, and σ_ν to this moment, the CHFS age–choice shares, and the regional income ratio. The universal transfer and inheritance distribution satisfy the fiscal and terminal-estate conditions. The calibrated $\alpha_p = 0.4907839$ yields homeowner receipt of 62.37257 percent, within the declared 0.5-percentage-point tolerance of the 62.78539 percent target.

CHIP measures retrospective housing assistance, whereas f is an unrestricted monetary gift received at age 1 in the model. The incidence target provides an empirical discipline for giving; it does not equate the timing, purpose, or amount of observed and modeled transfers. The contribution brackets do not identify exact RMB gifts. The remaining recipient–nonrecipient location–tenure comparisons are descriptive diagnostics, not additional calibration targets.

B.7 Calibration Map and Fit

Table B.3 summarizes how the empirical objects discipline the model. The 18 nonnormalized location–tenure constants match three independent shares per age; the taste-shock scale matches the regional income ratio; parental altruism matches assistance receipt among young homeowners; and the owner-space shares and construction shifters reproduce the four purchase-price and rent targets. The universal transfer balances the government budget, and the inheritance nodes close the terminal-estate fixed point. The Region A-renter versus Region B-owner productivity gap is an approximate target for the recurring rental friction, whose schedule is set separately. The remaining externally assigned parameters in Tables 2 and B.4 are not chosen to match these quantitative moments.

Table B.4 collects numerical and institutional parameters omitted from the compact main calibration table. I set the bequest shift to $\underline{e} = 10^{-6}$, ensuring that warm-glow utility remains finite at zero terminal estate.

Table B.3: Empirical Objects and Calibration Discipline

Object	Construction or target	Model discipline
Age–location–tenure shares	Weighted urban CHFS profiles, smoothed as in Table B.1	$\{\xi^g_{oB}, \xi^g_{rA}, \xi^g_{oA}\}_{g=1}^6$
Age earnings and permanent talent	Weighted log-earnings regression and mean-one Fréchet fit	$\{G_g\}$, θ_z, s_z, and the 13-point z grid
Regional earnings premium	Weighted age-adjusted log-earnings regression	$\mathrm{TFP}_A/\mathrm{TFP}_B$
Young rA–oB productivity gap	9.0% among urban household heads aged 20–29 (Table B.2)	Approximate target for $\{\kappa^{\mathrm{rent}}_{rA,g}\}_{g=1}^6$
Working-age A/B gross-income ratio	Equilibrium target of 2.0	σ_ν
Parental assistance among young homeowners	Unweighted CHIP 2018 incidence: $275/438 = 62.79\%$	α_p
Regional purchase prices and rents	Equations (B.2) and (B.3)	$\Omega_{A,0}, \Omega_{B,0}, \bar{I}_A, \bar{I}_B$
Government budget	Payments equal tax revenue in the stationary equilibrium	T
Terminal-estate distribution	Conditional mean estates within fixed quantile bins	Inheritance nodes

Notes: Positive-transfer incidence among young homeowners is targeted; unconditional incidence remains a model outcome. Housing-size moments and conditional productivity levels are also untargeted.

Table B.4: Additional Calibration and Institutional Parameters

Parameter	Explanation	Value	Source or role
θ_z	Fréchet shape	1.9266	CHFS earnings
s_z	Fréchet scale	0.5430	CHFS earnings
$\underline{e}$	Bequest shift	10^{-6}	Numerical regularization
N_b	Inheritance bins	5	Appendix A.3
$\{p^b_k\}_{k=1}^5$	Inheritance probabilities	$(0.25, 0.25, 0.25, 0.15, 0.10)$	Fixed quantile-bin widths
$\underline{h}$	Minimum housing size	10 m^2	25% of urban average

Notes: Monetary values are in units of 10,000 RMB. The inheritance nodes are updated to match the stationary terminal-estate distribution; mortgage payment and balance coefficients follow from the mortgage rate and remaining maturity.

Assigned preference parameters. The consumption–housing composite assigns equal CES weights, $\alpha_h = 0.5$. This is a utility weight rather than a fixed housing expenditure share. The curvature $\epsilon = -2$ implies an intratemporal elasticity of substitution of $1/(1-\epsilon) = 1/3$. Relative risk aversion is $\gamma = 2$, as in Mabille (2023). The annual discount factor is 0.95, within the range used in quantitative life-cycle housing models (Campbell and Cocco, 2003, Favilukis et al., 2017), giving $\beta = 0.95^{10} = 0.5987$ over a ten-year model period. The terminal-estate warm-glow weight is $\omega_b = 0.10$, following the bequest-motive specification in De Nardi (2004). Appendix D.2 varies consumption–housing curvature and parental altruism.

Financial contracts. Households save in a nonnegative liquid asset and cannot issue unsecured debt. Its annual net return is $r_s^{\text{ann}} = 1.75\%$, the one-year RMB deposit rate posted by the Bank of China during the sample period (Bank of China, 2015); the ten-year gross return is $R_s = (1.0175)^{10} = 1.1894$. Mortgages are fully amortizing at $r_m^{\text{ann}} = 5.01\%$, the weighted-average personal housing-loan rate reported by the PBC for 2017:Q3 (People's Bank of China, 2017). The loan-to-value ceiling is $\theta_{\text{LTV}} = 0.70$, requiring a down payment of at least 30 percent. The payment-to-income ceiling is $\theta_{\text{PTI}} = 0.50$ of gross labor income, following the Chinese regulatory limit documented by Fang et al. (2016).

The age-dependent maximum mortgage terms in Table B.5 allow 30 years for the first three age groups, 20 years at ages 50–59, 10 years at ages 60–69, and no new mortgage in the final retired stage. The 30-year maximum follows standard residential mortgage contracts in China (Bank of China, 2008). The payment and remaining-balance coefficients π_q and ϑ_q follow mechanically from the mortgage rate and remaining maturity; Appendix A.2 derives them and reports their values.

Table B.5 collects the full age-dependent schedules summarized in Table 2. Panel B reports the 18 location–tenure constants calibrated to the age-specific shares in Table 3.

Table B.5: Age-Dependent Calibration Schedules

Panel A. Earnings, mobility, and mortgage schedules

Age group	G_g	$\kappa_{A,g}^{\text{entry}}$	$\kappa_{rA,g}^{\text{rent}}$	K_g
20–29	0.844	4	2.0	3
30–39	1.000	5	2.0	3
40–49	0.968	6	2.5	3
50–59	0.911	7	3.0	2
60–69	0.670	8	3.0	1
70–79	0.000	9	3.0	0

Panel B. Location–tenure utility constants

Age group	ξ_{oB}^g	ξ_{rA}^g	ξ_{oA}^g
20–29	0.00558	-0.00097	0.00527
30–39	0.01915	0.00586	0.02185
40–49	0.01835	-0.00005	0.02000
50–59	0.01315	0.00001	0.01144
60–69	0.01127	0.00704	0.01820
70–79	0.01653	0.02150	0.04240

Notes: G_g is normalized to one at ages 30–39. Entry costs and recurring rental frictions are in units of 10,000 RMB; K_g is the maximum new mortgage term in ten-year blocks. The utility constant for renting in Region B is normalized to $\xi_{rB}^g = 0$ for every age group. The remaining constants are measured in flow-utility units. Table 3 reports the targeted choice shares and model fit.

All calibrated blocks, value functions, and the stationary distribution meet the toler-

ances in Appendix A. The maximum share error is 0.00855, the working-age A/B income ratio is 1.968, and the absolute budget and inheritance gaps are 4.1×10^{-4} and 5.0×10^{-4}. The homeowner-receipt error is -0.413 percentage points.

B.8 CHIP Descriptive Comparisons and Model Diagnostics

This comparison retains households with heads aged 20–29 to correspond to model age 1; Section 2 uses the broader 20–39 sample.

Table B.6 compares the CHIP location–tenure statistics with the baseline model's endogenous recipient and nonrecipient groups. In CHIP, receipt means reported historical parental assistance with buying or building housing; in the model, it means a positive, unrestricted monetary transfer at ages 20–29. The table therefore provides a descriptive comparison rather than an exact measurement-matched test.

The baseline reproduces higher overall ownership, higher ownership conditional on Region A residence, and lower Region A residence among recipients. However, its conditional-ownership gap is substantially smaller than the empirical gap, and it does not reproduce the positive empirical recipient gap in joint ownership and residence in A. This discrepancy remains a limitation of the baseline comparison. These within-economy group differences must not be interpreted as the model's equilibrium effects of eliminating transfers. The recipient–nonrecipient differences reported here are not calibration targets. The altruism parameter instead targets assistance receipt among young homeowners, as documented in Appendix B.6.

Table B.6: Location and Tenure by Assistance Receipt: CHIP and the Baseline Model

	CHIP		Model	
Outcome	No support	Positive support	No support	Positive support
Own, either region	52.9	89.6	47.9	57.4
Reside in A	54.2	40.7	40.0	31.5
Own conditional on A	40.1	80.8	36.1	41.4
Own and reside in A	21.8	32.9	14.4	13.0

Notes: All entries are percentages. CHIP uses unweighted urban households with heads aged 20–29 (308 nonrecipients; 307 recipients); current-dwelling tenure does not verify individual legal title. Survey assistance is historical parental help with buying or building housing, whereas model receipt is a positive unrestricted monetary gift at age 1 (20–29). Model entries use stationary state mass and choice probabilities from registered baseline0928 economy B, calibrated $\alpha_{\text{parent}} = 0.4908$; they are not sampling estimates. Conditional ownership equals $\Pr(oA)/\Pr(A)$. Free and other accommodation remain in the CHIP denominators. These descriptive group differences are not equilibrium effects of eliminating transfers. Source: CHIP 2018 urban household and person files; registered baseline0928 saved results; authorś calculations.

Appendix C Outcome Definitions, Decompositions, and Welfare

This appendix defines the measures used in Section 6. Differences are $B - B^0$.

C.1 Residence–Ownership and Effective-Labor Decompositions

For economy $s \in \{B, B^0\}$, let $A_s = \Pr(A)_s$ and $q_s = \Pr(\text{own} \mid A)_s$. Since $\Pr(oA)_s = A_s q_s$, the exact symmetric decomposition is

$$\Delta oA = \underbrace{\frac{q_B + q_{B^0}}{2}(A_B - A_{B^0})}_{\mathcal{S}\text{: spatial component}} + \underbrace{\frac{A_B + A_{B^0}}{2}(q_B - q_{B^0})}_{\mathcal{T}\text{: tenure component}}.$$

The spatial component changes residence at cross-equilibrium average conditional ownership; the tenure component changes conditional ownership at average residence. Table 5 reports the tenure share $100\mathcal{T}/\Delta oA$ when $\Delta oA \neq 0$.

Let $\mu_A^{g,s}$ be the measure of age-g households in A, including stationary age weights of $1/6$ across both regions. Working-age mass, mean age-adjusted productivity, and effective labor are

$$M_A^s = \sum_{g=1}^{5} \int d\mu_A^{g,s}, \qquad \overline{Gz}_A^s = \frac{\sum_{g=1}^{5} \int G_g z \, d\mu_A^{g,s}}{M_A^s}, \qquad L_A^s = M_A^s \overline{Gz}_A^s. \tag{C.1}$$

The corresponding decomposition is

$$\Delta L_A^{\text{mass}} = \frac{\overline{Gz}_A^B + \overline{Gz}_A^{B^0}}{2}(M_A^B - M_A^{B^0}), \tag{C.2}$$

$$\Delta L_A^{\text{comp}} = \frac{M_A^B + M_A^{B^0}}{2}(\overline{Gz}_A^B - \overline{Gz}_A^{B^0}), \tag{C.3}$$

$$L_A^B - L_A^{B^0} = \Delta L_A^{\text{mass}} + \Delta L_A^{\text{comp}}. \tag{C.4}$$

The first term changes worker mass at average productivity; the second changes average productivity at average worker mass. These are accounting components, not separately identified causal effects. Working ages have total mass $5/6$, so their Region A residence share is $M_A^s/(5/6)$. The same construction applies to Region B; output is $Y_\ell^s = \text{TFP}_\ell L_\ell^s$, with $Y^s = Y_A^s + Y_B^s$.

C.2 Equity, Net Worth, and Parental-Wealth Groups

For household i in economy s, end-of-stage home equity and net worth are

$$HE_i^s = \mathbb{1}\{i \text{ owns}\}(P_{\ell_i}^s h_i^s - m_i'^{,s}), \qquad NW_i^s = x_i'^{,s} + HE_i^s.$$

Renters enter home-equity statistics at zero. Housing is valued at each economy's equilibrium price and debt is measured after amortization. Statistics use stationary-mass weights normalized within the reported group. Labor-income inequality pools ages 1–5; disposable-income, consumption–housing, and net-worth measures pool all six ages unless an age group is specified. Disposable income and the composite follow equations (16) and (2). The age-1 talent gap is the mean outcome of the top 20 percent minus that of the bottom 80 percent.

Parental wealth is measured before age-4 giving as

$$NW_{p,i}^{\text{beg},s} = x_{p,i}^s + \mathbb{1}\{i \text{ owns}\}[(1-\delta_h)P_{\ell_{p,i}}^s h_{p,i}^s - m_{p,i}^s].$$

Weighted quartile cutoffs are calculated in B and applied unchanged in both economies. The Q4–Q1 gap compares weighted mean linked-child outcomes across these parental groups, integrating over child-talent draws and age-1 choice probabilities.

C.3 Self-Lifetime Welfare and Consumption-Equivalent Variation

For a household currently in age group g, self-lifetime welfare is

$$\mathcal{W}_g^s = \mathbb{E}_s\left[\sum_{a=g}^{6} \beta^{a-g}\left\{u^a(c_a, h_a, j_a) + \mathbb{1}\{a=6\}\omega_b \frac{(e_6 + \underline{e})^{1-\gamma}}{1-\gamma}\right\}\right]. \tag{C.5}$$

The deterministic flow utility u^a in equation (1) includes location–tenure utility; e_6 is the terminal estate in equation (24). The criterion retains terminal warm glow but excludes logit taste utility and the altruistic child-value term. Excluding the latter avoids counting the linked child's welfare in both the parent's and child's measures.

The consumption–housing CEV solves

$$\mathcal{W}_g^{B^0}\big((1+\lambda_g^{c+h})c, (1+\lambda_g^{c+h})h; e_6\big) = \mathcal{W}_g^B. \tag{C.6}$$

Consumption and housing services are scaled along the fixed B^0 allocation through the remaining lifetime. Location–tenure choices, their deterministic utility constants, and terminal estates remain unchanged. Positive CEVs indicate compensation required in B^0 to

match B; negative CEVs indicate higher welfare in B^0. The consumption-only diagnostic scales consumption alone, leaving housing and estates unchanged, under the same CES utility and weighting conventions.

Expectations use each economy's stationary distribution and policy probabilities, counting choice probabilities once and normalizing within each reported age–talent group. Talent groups use the same percentile cutoffs in both economies. For all living households, a single compensation $\lambda_{\text{all}}^{c+h}$ solves

$$\sum_{g=1}^{6} \frac{1}{6} \mathcal{W}_g^{B^0}\big((1+\lambda_{\text{all}}^{c+h})c, (1+\lambda_{\text{all}}^{c+h})h; e_6\big) = \sum_{g=1}^{6} \frac{1}{6} \mathcal{W}_g^{B}.$$

Thus, stationary age weights are applied to welfare before solving for common compensation; the all-living CEV is not an average of age-specific CEVs. Older households' welfare differences combine earlier receipt, accumulated positions, current giving, and the equilibrium environment, rather than isolating the current cost of giving.

Appendix D Sensitivity Analysis

This appendix tests whether the distinction between spatial access and ownership survives alternative preferences, parental altruism, housing-supply conditions, mobility frictions, rental access, and intergenerational talent persistence. It then evaluates numerical accuracy by refining one state-space dimension at a time.

D.1 Sensitivity Protocol

Each economic sensitivity is evaluated as a pair of stationary equilibria. When a primitive changes the baseline household problem, I recalibrate the 18 age-specific location–tenure constants, T, σ_ν, and α_p to the same 24 age–choice shares, the same Region A-to-Region B income ratio, and the CHIP homeowner-receipt target. In the explicit $\alpha_p = 0.30$ and 0.60 exercises, α_p is fixed and homeowner receipt is reported without being targeted. Except for the predetermined parameter under examination, all other predetermined parameters remain at their benchmark values; prices and rents remain fixed at their calibration targets, and inheritance is updated endogenously. The housing-supply and owner–rental conversion cases do not change the calibrated baseline allocation and therefore reuse the benchmark transfer equilibrium.

An accepted transfer equilibrium must have a maximum age–choice share error no larger than 0.02, an income-ratio error no larger than 2 percent, a homeowner-receipt error no larger than 0.5 percentage points when targeted, a government-budget residual no

larger than 10^{-3}, an inheritance gap no larger than 10^{-3}, and converged value functions and stationary distribution. For each accepted transfer economy, I back out housing supply from its demand and solve a new no-transfer equilibrium with $f = 0$, jointly updating prices, rents, T, and the inheritance distribution. The nominal housing-market tolerance is 10^{-3}. Small persistent oscillations may be accepted up to 1.5×10^{-3}; the separately recorded exception for fixed owner–rental conversion uses the same upper limit. The inelastic-supply no-transfer case is separately accepted with a housing gap of 1.71×10^{-3}; it does not meet the nominal criterion. Fiscal, inheritance, and inner-solution criteria are unchanged. The tighter GE check retains its 5×10^{-4} criterion. Only accepted cases enter the final comparisons.

All effects below are transfer equilibrium minus no-transfer equilibrium. The primary mechanism is *preserved* when transfers raise the age-1 Region A share and raise the age-1 oA share by still more. It is *mixed* when only one condition holds and *overturned* when neither holds. The tenure share uses the decomposition in Appendix C.1. The tables also report the age-1 home-equity effect and the aggregate-output effect normalized by the benchmark transfer effect. Separate talent-group diagnostics classify low- and middle-talent access gains as broad or uneven. A verdict is marked fragile when its defining margin is smaller than the 0.218 percentage-point numerical envelope established in Appendix D.6.

D.2 Preferences and Parental Altruism

The preference tests vary the CES curvature from its benchmark value $\epsilon = -2$ to -3 and -1. The altruism tests set α_p to 0.30 and 0.60. Table D.1 reports the resulting transfer effects.

Table D.1: Sensitivity to Preferences and Parental Altruism

Specification	$\Delta \Pr(A)$	$\Delta \Pr(oA)$	Tenure share	ΔHE_1	$\Delta Y / \Delta Y_{\text{bench}}$	Mechanism
Benchmark	1.177	4.295	90.9%	8.34	1.00	Preserved
$\epsilon = -3$	1.288	4.425	90.5%	8.14	1.10	Preserved
$\epsilon = -1$	1.332	4.096	89.6%	9.29	0.87	Preserved
$\alpha_p = 0.30$	0.582	2.939	93.9%	5.65	0.69	Preserved
$\alpha_p = 0.60$	1.479	4.976	89.8%	9.89	1.14	Preserved

Notes: Probability effects are percentage points. Tenure share is in percent; ΔHE_1 is in units of 10,000 RMB. The output column divides each aggregate-output effect by the benchmark effect.

Both curvature changes preserve the mechanism. Raising α_p increases the oA, home-equity, and welfare effects, whereas lowering it attenuates them.

D.3 Intergenerational Talent Persistence

The benchmark draws a young adult's talent independently of parental talent. I instead use the parent–child transition matrix

$$\Pi(r) = (1 - r)\mathbf{1}p' + rI,$$

where p is the stationary talent distribution and r governs persistence. This specification preserves the marginal talent distribution. I consider $r = 0.30$ and $r = 0.50$, recalibrate the baseline, and resolve the no-transfer equilibrium in each case. Table D.2 reports the resulting transfer effects.

Table D.2: Sensitivity to Intergenerational Talent Persistence

Specification	$\Delta \Pr(A)$	$\Delta \Pr(oA)$	Tenure share	ΔHE_1	$\Delta Y/\Delta Y_{\text{bench}}$	Mechanism
Independent ($r = 0$)	1.177	4.295	90.9%	8.34	1.00	Preserved
Moderate ($r = 0.30$)	0.923	3.425	91.4%	6.63	0.74	Preserved
High ($r = 0.50$)	0.807	2.866	91.2%	5.58	0.61	Preserved

Notes: Units and columns follow Table D.1. All three specifications satisfy the calibration and equilibrium-convergence criteria.

Persistence attenuates the spatial, ownership, home-equity, and output effects but leaves their ordering unchanged. Even at $r = 0.50$, transfers raise Region A residence by 0.81 percentage points and oA by 2.87 points, with 91 percent of the ownership response operating through tenure. The central mechanism therefore does not depend on independent parent and child talent draws.

D.4 Housing Supply and Owner–Rental Conversion

These tests vary the housing-supply elasticities around the benchmark $(\rho_A, \rho_B) = (0.65, 0.95)$ and the owner–rental conversion elasticities around $(\chi_A, \chi_B) = (1, 1)$. They change the GE response to removing transfers without changing the calibrated transfer allocation. Table D.3 reports the results.

The ownership effect ranges from 4.04 to 4.38 percentage points throughout these tests, and the tenure margin explains 89.5–92.5 percent of it. Supply elasticity and conversion flexibility change equilibrium prices and rents, but they do not materially alter the distinction between Region A access and ownership.

Table D.3: Sensitivity to Housing Supply and Owner–Rental Conversion

Specification	$\Delta \Pr(A)$	$\Delta \Pr(oA)$	Tenure share	ΔHE_1	$\Delta Y/\Delta Y_{\text{bench}}$	Mechanism
Benchmark	1.177	4.295	90.9%	8.34	1.00	Preserved
$(\rho_A, \rho_B) = (0.45, 0.70)$	1.233	4.277	90.4%	8.34	0.96	Preserved
$(\rho_A, \rho_B) = (1.00, 1.50)$	1.042	4.315	92.0%	8.27	0.98	Preserved
$(\rho_A, \rho_B) = (1.80, 2.70)$	0.983	4.352	92.5%	8.31	0.96	Preserved
$(\chi_A, \chi_B) = (0, 0)$	1.260	4.038	89.5%	8.20	0.93	Preserved
$(\chi_A, \chi_B) = (2, 2)$	1.111	4.382	91.6%	8.36	1.01	Preserved

Notes: Units and columns follow Table D.1. In the ρ rows, $\chi_A = \chi_B = 1$; in the χ rows, $(\rho_A, \rho_B) = (0.65, 0.95)$.

D.5 Mobility Costs and Region A Rental Frictions

The mobility tests multiply the benchmark Region A entry-cost schedule by 0.5 and 1.5. The rental-access tests eliminate or double the recurring Region A rental-friction schedule. These assumptions have the largest effect on the spatial margin because they directly change the cost of reaching Region A through renting. The zero-friction specification is also a mechanism-removal test of the benchmark sorting discipline, not an alternative estimate of the institutional cost. Table D.4 summarizes these comparisons.

Table D.4: Sensitivity to Mobility Costs and Rental Frictions

Specification	$\Delta \Pr(A)$	$\Delta \Pr(oA)$	Tenure share	ΔHE_1	$\Delta Y/\Delta Y_{\text{bench}}$	Mechanism
Benchmark	1.177	4.295	90.9%	8.34	1.00	Preserved
Entry costs $\times 0.5$	-0.412	4.079	103.0%	8.33	0.92	Mixed
Entry costs $\times 1.5$	2.183	4.236	83.0%	7.97	1.12	Preserved
Rental friction $= 0$	-1.231	4.370	108.4%	8.64	0.70	Mixed
Rental friction $\times 2$	3.189	4.048	73.3%	7.94	1.30	Preserved

Notes: Units and columns follow Table D.1. The benchmark entry-cost schedule is $(4, 5, 6, 7, 8, 9)$; the benchmark rental-friction schedule is $(2, 2, 2.5, 3, 3, 3)$.

Lower entry costs and the removal of rental frictions reverse the residence effect: transfers reduce age-1 Region A residence by 0.41 and 1.23 percentage points, respectively. In both cases, transfers still raise oA by 4.08 and 4.37 percentage points, respectively. The tenure share exceeds 100 percent because a negative spatial component partly offsets the positive tenure component. Doubling the rental friction instead increases the spatial effect to 3.19 percentage points. Thus, the ownership result is robust, while the sign and size of the spatial-access effect depend on the cost of renting and moving to Region A.

The rental-friction reversal reflects the competing alternatives emphasized in Section 5.2. Parental support finances ownership in Region A, but modest transfers also

make lower-cost ownership in Region B attractive. The recurring rental friction weakens renting as a substitute for transfer-financed ownership in Region A. Removing it makes Region A accessible through renting with less reliance on family resources, reducing the spatial-access benefit of transfers while preserving their ability to finance purchases in Region B. In the recalibrated zero-friction economy, the resulting balance reverses the net residence effect even though ownership in Region A still rises. These equilibrium comparisons incorporate recalibration and housing-market adjustment.

The CHFS evidence on average talent sorting provides an empirical reason to retain a positive rental friction in the benchmark. Removing the friction reduces the mean-talent gap between young Region A renters and Region B owners from 11.4 to 4.8 percent, compared with 9.0 percent among young household heads in CHFS. The positive ordering survives, but its strength is substantially weaker. The benchmark profile targets the direction and approximate size of the household-head gap; it does not impose an exact match.

D.6 Numerical Accuracy

The stationary distribution is deterministic, so simulation-seed tests are not applicable. I instead refine the transfer, talent, liquid-asset, housing, and inheritance grids separately and tighten the GE tolerance. Refining one dimension at a time avoids an infeasible simultaneous expansion of the state space. Table D.5 reports the resulting accuracy checks.

Table D.5: Numerical Accuracy of the Transfer Mechanism

Variant	n_z	n_f	n_x	n_h	n_b	$\Delta \Pr(A)$	$\Delta \Pr(oA)$	$\lvert d_A \rvert$	$\lvert d_M \rvert$
Benchmark	13	13	10	13	5	1.177	4.295	0.000	0.000
Denser transfer grid	13	22	10	13	5	1.145	4.282	0.032	0.019
22-point talent grid	22	13	10	13	5	1.379	4.280	0.202	0.218
13-point asset grid	13	13	13	13	5	1.156	4.316	0.021	0.041
25-point housing grid	13	13	10	25	5	1.337	4.359	0.160	0.096
Seven inheritance bins	13	13	10	13	7	1.168	4.293	0.009	0.006
Nine inheritance bins	13	13	10	13	9	1.167	4.294	0.010	0.009
GE tolerance 5×10^{-4}	13	13	10	13	5	1.167	4.291	0.010	0.006

Notes: Probability effects and discrepancies are percentage points. n_b is the number of inheritance bins. d_A is the change in $\Delta \Pr(A)$ relative to the benchmark; d_M is the change in the tenure margin $\Delta \Pr(oA) - \Delta \Pr(A)$. All variants satisfy their applicable acceptance criteria. The denser transfer grid, 22-point talent grid, and 25-point housing grid trigger the conditional recalibration rule; the remaining checks retain fitted preference parameters. Discrepancies include these prescribed recalibrations.

The largest numerical discrepancy is 0.218 percentage points, generated by the 22-point talent grid. Every refinement preserves a positive Region A effect and an ownership

effect that is substantially larger. The aggregate-output effect remains positive across all seven checks, ranging from 0.225 to 0.239 percent. These discrepancies measure observed numerical sensitivity, including the prescribed recalibrations, rather than a formal error bound.

The direction of the small Region A rent effect is not stable across grids: transfers lower rent by 0.49 percent in the benchmark but raise it by 0.10 percent with the refined housing grid. The benchmark rental-expense benefit should therefore not be interpreted as a result that holds across all numerical checks.

Across the 16 accepted economic specifications, 14 preserve the full mechanism and 2 are mixed; none overturns both conditions. The mixed cases are lower entry costs and zero rental friction, and both spatial reversals exceed the numerical envelope. Every accepted specification also produces positive age-1 home-equity and aggregate-output effects, and the all-living welfare effect remains positive, ranging from 6.67 to 12.90 percent.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT and Codex (OpenAI) for language polishing and manuscript organization. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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